$x^2+y^2+z^2-2x+2z=0$

1

$4+1+\cancel{1}\,2-2$ Outbreak 50

In the euclidean 3-demensional space with a fixed cartesian system, let us consider the sphere S defined by the equation.

$x^2+y^2+z^2-2x+2z=0$

a) S intersects the line given by $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ t \\ 0 \end{pmatrix}$ in two distinct points

b) S passes through the point $(1,-1,1)$

c) the center of S has coordinates $(2,1,-1)$

d) S has radius $\sqrt{2}$.

In the Euclidean 3-space let us consider the points with coordinates $A=(0,0,a)$, $B=(2,-2,a)$ $C=(2,2,a)$ with $a \in R$

The area of the triangle ABC is equal to

a) 4 for every $a \in R$

b) 2 for exactly one $a \in R$

c) $\frac{a^2}{2}$ for every $a \in R$

d) $\frac{a^2}{2}$ for exactly one $a \in R$

Let $V = \{(x,y,z) \in R^3 : x+y-z=0\}$ and let $f: R^3 \to R^3$ be an endomorphism such that $V = \ker(f)$ and 1 is an eigenvalue of f

a) The characteristic polynomial of f can be $t(t-1)^2$

b) f has three distinct eigenvalues

c) $\dim(Im(f)) = 2$

d) f is simple.

Let us consider the quadratic form defined by

$q(x,y) = (x,y)\,A\begin{pmatrix} x \\ y \end{pmatrix} = -2x^2 + 4xy + y^2$,

where A is a 2×2 symmetric matrix

ⓐ There exists $(a,b) \neq (0,0)$ such that $q(a,b) = 0$

ⓑ if $xy \neq 0$ then $q(x,y) > 0$

ⓒ the characteristics polynomial of A equal to $t^2 + t + 6$

ⓓ the matrix A has one eigenvalue equal to 0 (zero)

Let $\cdot$ denote the usual matrix multiplication. if

$A = [1\ 1\ 1] \qquad B = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$

Ⓐ $B\cdot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Ⓑ $B\cdot A = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \end{bmatrix}$

Ⓒ $A\cdot B = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$

Ⓓ $B\cdot A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$

Let S be a linear system of 4 equations and 3 variables, and let $M = (A|B)$ be the complete matrix associated to S

if $\det(M) \neq 0$, which one of the following statements is certainly true

ⓐ S admits infinite solutions

ⓑ S does not have any solutions

ⓒ S admits only one solution

ⓓ the zero vector is a solution of S

1) $x^2+y^2-2x-6y+10=0$.

$A=1 \quad B=0 \quad C=1 \quad D=-2 \quad E=-6 \quad F=10$.

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -3 \\ -1 & -3 & 10 \end{pmatrix} = 10-1-9=0$$

$A=(0,a,0) \quad B=(3,a,-3) \quad C=(3,a,3)$.

$AB=(3,0,-3) \quad AC=(3,0,3)$.

$\cos\alpha = \dfrac{3\cdot 3-9}{\dots} = 0. \quad S=\dfrac{1}{2}\sqrt{18}\cdot\sqrt{18} = \dfrac{2\sqrt{2\cdot 9}}{2} = 3\sqrt{2}$

$\alpha = 90$. $\quad$ 18

$2 \quad 1 \quad\quad 1 \quad 3 \quad \begin{bmatrix} 5 \\ 5 \end{bmatrix} (1,1,1) =$

2×3

$x^2+y^2+z^2-2x+2z=0$.

$x=1 \quad y=0 \quad z=-1 \quad (1,0,-1)$

$$\begin{pmatrix} -2 & 2 \\ 2 & 1 \end{pmatrix} = (\lambda+2)(\lambda-1)-4=0.$$

$\lambda^2+\lambda-2-4=0$.

$\lambda^2+\lambda-6=0$.

$\lambda=-3 \quad \lambda=2$.

$-2\cdot 9+$

$-8-4$

$-2x^2+4xy+y^2$.

$-2\cdot 1+4\cdot 1\cdot 1+1$

$-2\cdot 1+4+1$

$r: \begin{cases} x - y + 2z = 0. \\ x + y + z = 2. \end{cases}$ $\quad$ $s: \begin{cases} -x + y + 3z = 0. \\ 2x - y + 2z = 1. \end{cases}$

$r \left(\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 2 \end{array}\right)$ $\quad$ $z = t$, $y = 2 + t$, $x = t.$

t

$t(1, 1, 1) + 2(0, 1, 0).$

$\left(\begin{array}{ccc|c} -1 & 1 & 3 & 0 \\ 2 & -1 & 2 & 1 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} -1 & 1 & 3 & 0 \\ 0 & 1 & 8 & 1 \end{array}\right)$ $\quad$ $z = t.$ $y = 1 - 8t.$

t

$-x + 1 - 8t + 3t = 0.$

$x = 1 - 5t.$

$t \begin{pmatrix} -5 \\ -8 \\ 1 \end{pmatrix} + (1, 1, 0).$

$y - z = 0.$

$V = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$ $\quad$ $W = \begin{pmatrix} 0 & k \\ h & d \end{pmatrix}$

$V = \{y - z\}$ $\quad$ $V = \ker(f).$ $\quad$ eigen value $-1 = f$

$V = \{(x, y, z) \in \mathbb{R}^2 : y - z = 0$ $\quad$ $f: \mathbb{R}^3 \to \mathbb{R}^3$

$V = \ker(f).$ $\quad$ -1 eigen value of $f.$

$\begin{cases} x - y + 2z = 0. \\ . \end{cases}$ $\quad$ $\left(\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 1 & 1 & 1 & 2 \end{array}\right) \Rightarrow$

1) $V = \{(x,y,z) \in R^3 : 2x+y-3z=0\}$, $f: R^3 \to R^3$, $V = \ker(f)$, 0 eigenvalue of f 3

2) $S: x^2+y^2+z^2-2y+2z=0$

3) $\begin{cases} x-y+2z=0 \\ 2x+y+z=3 \end{cases}$ $\quad$ $\begin{cases} x+2y-z=3 \\ x+z=1 \end{cases}$

4) S: 4 equations, 3 variables, $M=(A|B)$ matrix, $\det(M)=0$

6) $C: -4x^2+y^2+2y+y=0$

7) $q(x,y) = (x,y)\, A \begin{pmatrix} x \\ y \end{pmatrix} = -2x^2+4xy+y^2$

$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

9) $f: R^{3,2} \to R^{2,2}$, $f(X) = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix} \cdot X$, $\forall X \in R^{3,2}$

10) $V\left(\begin{pmatrix} 0 & a \\ b & c \end{pmatrix} \mid a,b,c \in R\right)$, $W = \left(\begin{pmatrix} d & e \\ e & 0 \end{pmatrix} \mid d,e \in R\right)$

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the followin subsets of $\mathbb{R}_3[x]$ is a vector subspace?

(a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

(b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗

(c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.

(d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let $\mathcal{C}$ be the conic defined by the equation

$$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0.$$

Which of the following statements is correct?

(a) $\mathcal{C}$ is a hyperbola.

(b) $\mathcal{C}$ is degenerate. ✗

(c) $\mathcal{C}$ is a parabola.

(d) $\mathcal{C}$ is an ellipse.

Risposta errata.

The correct answer is: $\mathcal{C}$ is a parabola.

(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$ ✓

Risposta corretta.

The correct answer is: $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $\mathbb{R}_3[x]$ is a vector subspace?

(a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

(b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗

(c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.

(d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let $\mathcal{C}$ be the conic defined by the equation

$$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0.$$

Which of the following statements is correct?

(a) $\mathcal{C}$ is a hyperbola.

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Flag question

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & -1 & 2 \end{pmatrix}$$

Find the correct statement.

(a) $A^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & -1 & -3 \\ 1 & -1 & -1 \end{pmatrix}$

(b) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(c) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 2 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$ ✓

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A) = 0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x, y, z) = (-x + z, -2y, x - z)$.

Which of the following statements is correct?

(a) All the eigenspaces of f have dimension 1.

(b) The image of f has dimension 1.

(c) f is simple.

(d) f is an isomorphism.

✗

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ \ldots \end{pmatrix}$$

(a) There exists a unique plane containing the point $(2,2,0)$ and orthogonal to r.

(b) There exists a unique plane containing the point $(2,2,0)$ and parallel to r.

(c) There exists a unique plane containing r and the point $(2,2,0)$.
✗

(d) There exists a unique plane containing r and orthogonal to the vector $\vec{v} = (1,-1,1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2,2,0)$ and orthogonal to r.

Question 6

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zer
and find the correct statement.

(a) $\det(A) = 0$.

(b) $\mathrm{rank}(A) = n$. ✗

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A) = 0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x,y,z) = (-x+z, -2y, x-z)$.

Which of the following statements is correct?

100.00

Flag question

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(d) C_1 and C_2 coincide. ✗

Risposta errata.

The correct answer is: $R_2 > R_1$.

Question 5

Incorrect

Mark -15.00 out of 100.00

Flag question

Let us consider the line

$$r: \begin{cases} x = 2 + t \\ y = 2 - t \\ z = t \end{cases}$$

Which of the following statements is true?

(a) There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to $\boldsymbol{r}$.

(b) There exists a unique plane containing the point $(2, 2, 0)$ and parallel to $\boldsymbol{r}$.

(c) There exists a unique plane containing $\boldsymbol{r}$ and the point $(2, 2, 0)$.
✗

(d) There exists a unique plane containing $\boldsymbol{r}$ and orthogonal to the vector $\vec{v} = (1, -1, 1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to $\boldsymbol{r}$.

Question 3

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

(c) 0 is not an eigenvalue of A.

(d) None of the other statements is correct.

Risposta errata.

The correct answer is: A is not positive or negative definite.

estion 4

orrect

k -15.00 out of

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lag question

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1 : x + y + z = 2$ $\pi_2 : x + y + z = 0$ let us denote by R the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(d) C_1 and C_2 coincide. ✗

Risposta errata.

The correct answer is: $R_2 > R_1$.

n 5

Let us consider the line

HELP CENTER

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t-3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

Question 3

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

(c) 0 is not an eigenvalue of A.

(d) None of the other statements is correct.

Risposta errata.

The correct answer is: A is not positive or negative definite.

stion 4

rect

-15.00 out of

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ag question

Given the sphere $S: x^2+y^2+z^2-2x+2y=7$, and the planes $\pi_1: x+y+z=2$ $\pi_2: x+y+z=0$ let us deno

the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(c) $\dim(\text{Ker } f) = 2$.

(d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\text{Ker } f) = 1$.

Question 2

Incorrect

Mark -15.00 out of 100.00

Flag question

For all $t \in \mathbb{R}$, consider the points

$$P = (1, 2, 3), \quad Q = (0, 0, t), \quad R = (1, -1, t), \quad S = (1, 1, t).$$

Which of the following statements is true?

(a) Points P, Q, R, S are vertices of a parallelogram for all values of t.

(b) Points P, Q, R, S are coplanar for a unique value of t.

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

uestion 3

orrect

rk -15.00 out of

0.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

State	Finished
Completed on	Tuesday, 18 January 2022, 9:02 AM
Time taken	1 hour
Marks	-35.00/1000.00
Grade	**-1.16** out of 33.00 (**-4**%)

stion 1
rrect
: -15.00 out of
00
lag question

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f : \mathbb{R}_2[x] \to \mathbb{R}^3$ be a linear map such that

$$f(x^2) = (0,0,1), \quad f(x) = (0,1,0), \quad f(1) = (0,3,3).$$

Which one of the following statements is true?

(a) $f(3x-1) = (0,0,3)$. ✗

(b) $\dim(\operatorname{Ker} f) = 1$.

(c) $\dim(\operatorname{Ker} f) = 2$.

(d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\operatorname{Ker} f) = 1$.

stion 2
rect

For all $t \in \mathbb{R}$, consider the points

$P = (1, 2, 3)$ $Q = (0, 0, t)$ $R = (1, \ldots 1, t)$ $S = (1, 1, t)$

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...r algebra and geometry 1 - TTPU

Linear algebra and geom...

...ZIONE QUIZ

Domanda 2

Risposta non ancora data

Punteggio max 100,00

Contrassegna domanda

Let Π be the plane in $\mathbb{R}^3$ defined by the equation $x + 2z = 0$. Let P be the point with coordinates ...

Which of the following statements is true?

- (a) The orthogonal projection of P on the plane Π is the point having coordinates $(0, 1, 0)$
- (b) The orthogonal projection of P on the plane Π is the point having coordinates $(2, 0, -1)$
- (c) The orthogonal projection of P on the plane Π is the point having coordinates $(-2, 0, 1)$
- (d) The orthogonal projection of P on the plane Π is the point having coordinates $(1, 0, 2)$

5 6 7 8 9

...uccessivo

Precedente Successivo

Let Π be the plane $\mathbb{R}^3$

$3x^2 + y^2 + 2x + y = 0$

$$A_k = \begin{pmatrix} 4 & 3 & 1 \\ 8 & 1 & 0 \\ 1 & k & 3k-1 \end{pmatrix}$$

$x - y - z = k + k^2$

$x + z = k$

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NAVIGAZIONE QUIZ

SHAHRIYOR
SHARIPOV
U12132

1 2 3 4 5 6 7 8 9
10

Precedente Successivo

Tempo rimasto **0:41:22**

Domanda 9

Risposta non ancora data

Punteggio max 100,00

Contrassegna domanda

For $k \in \mathbb{R}$ let U_k be the subset of $\mathbb{R}^3$ given by the solutions to the system

$$\begin{cases} x-y-z=k+k^2 \\ x+z=k \end{cases}$$

Which of the following statements is true?

- (a) U_k is a vector subspace for every $k \in \mathbb{R}$
- (b) U_k is not a vector subspace for every $k \in \mathbb{R}$
- (c) There exists $k \in \mathbb{R}$ such that U_k is not a vector subspace.
- (d) There exists $k \in \mathbb{R}$ such that $U_k = \emptyset$

Precedente Successivo

SAMSUNG

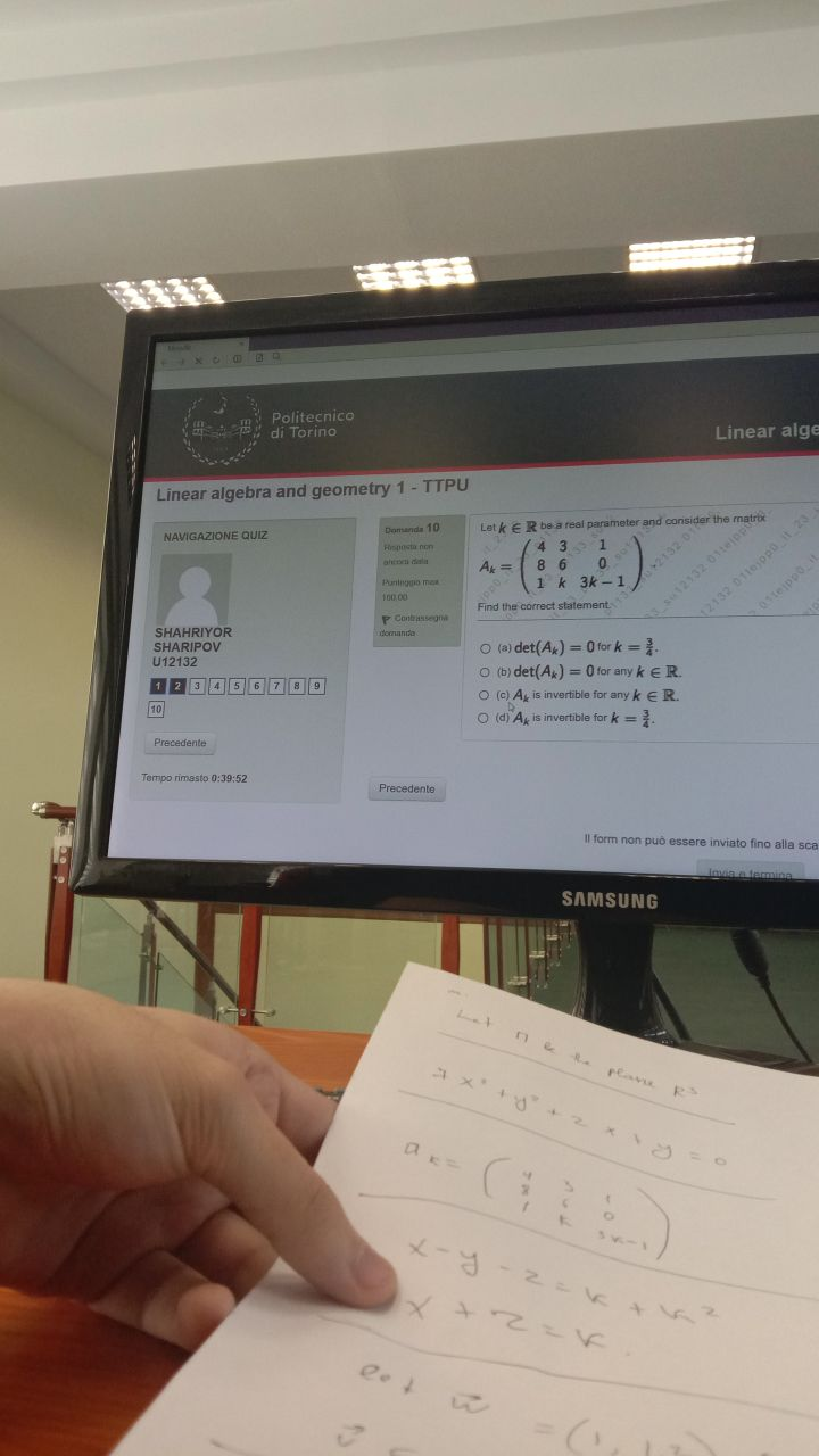
Let π be the plane R^3

$x^2 + y^2 + 2x + y = 0$

$a_k =$

$x - y - z = k + k^2$

$x + z = k$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Precedente
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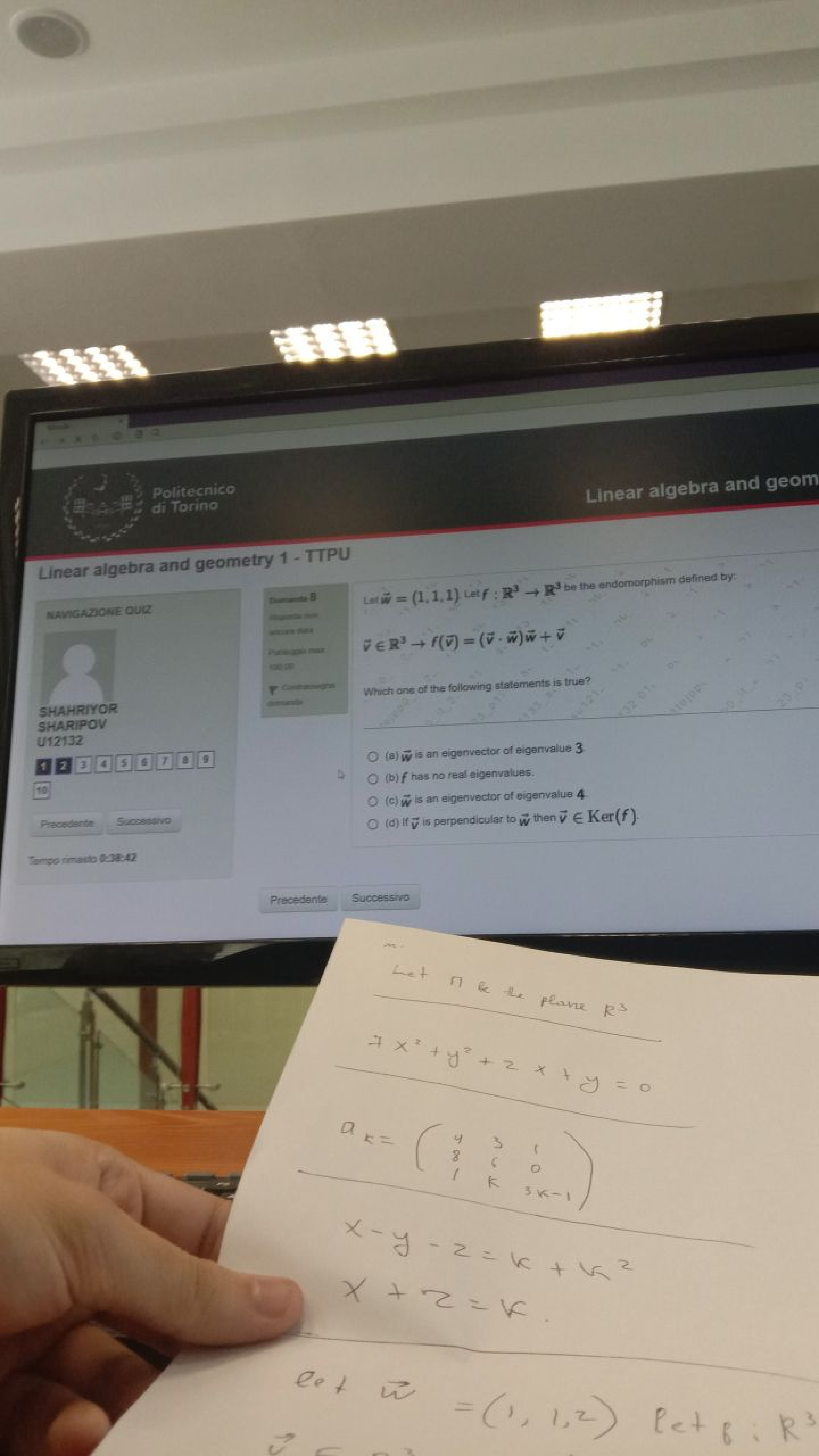
Domanda 10
Risposta non ancora data
Punteggio max 100,00
Contrassegna domanda
Let $k \in \mathbb{R}$ be a real parameter and consider the matrix
$A_k = \begin{pmatrix} 4 & 3 & 1 \\ 8 & 6 & 0 \\ 1 & k & 3k-1 \end{pmatrix}$
Find the correct statement.
(a) $\det(A_k) = 0$ for $k = \frac{3}{4}$.
(b) $\det(A_k) = 0$ for any $k \in \mathbb{R}$.
(c) A_k is invertible for any $k \in \mathbb{R}$.
(d) A_k is invertible for $k = \frac{3}{4}$.
Precedente
Il form non può essere inviato fino alla sca
Invia e termina
SAMSUNG
Let Π be the plane $\mathbb{R}^3$
$7x^2 + y^2 + 2x + y = 0$
$A_k = \begin{pmatrix} 4 & 3 & 1 \\ 8 & 6 & 0 \\ 1 & k & 3k-1 \end{pmatrix}$
$x - y - z = k + k^2$
$x + z = k$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Precedente Successivo
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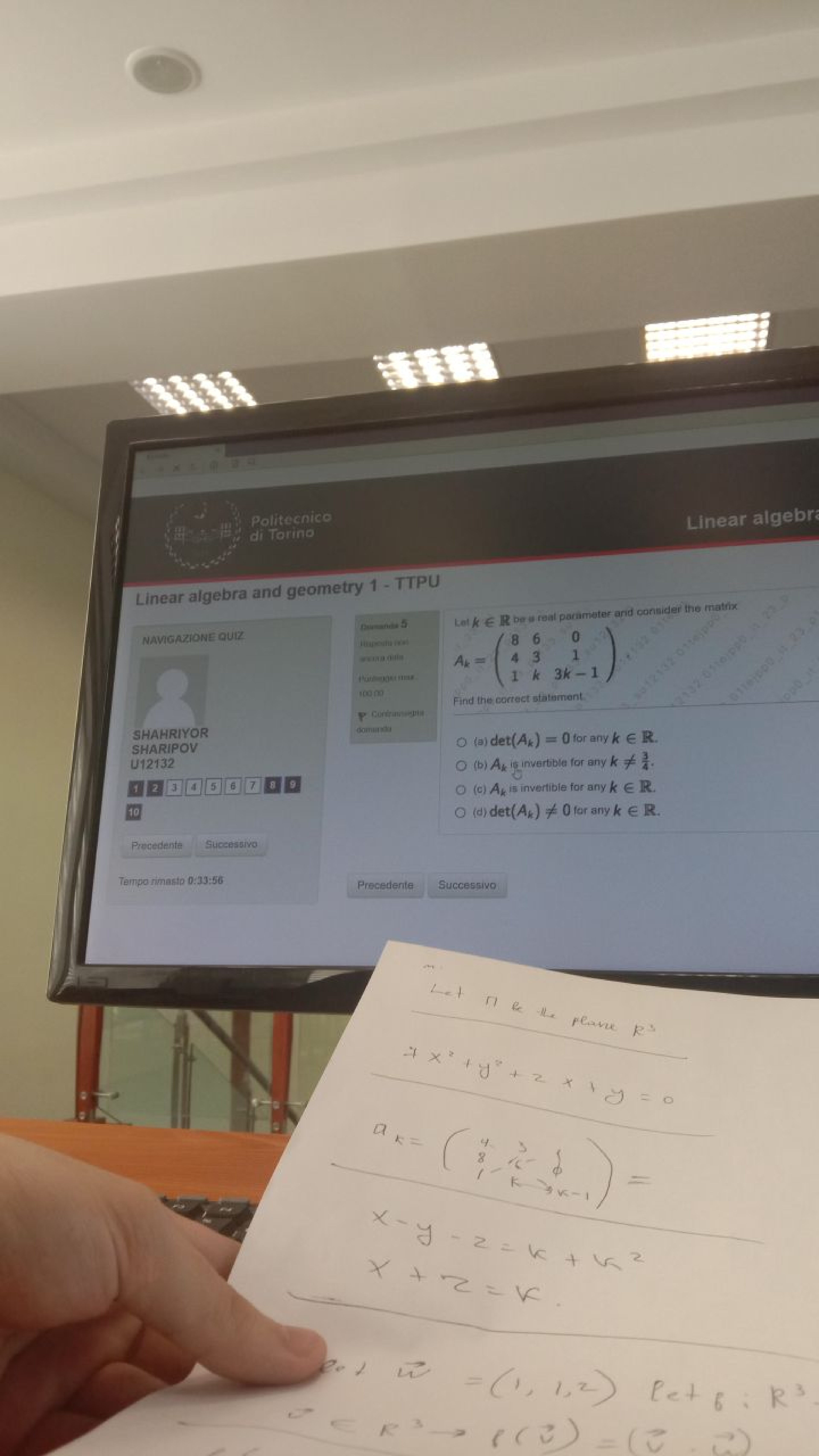
Domanda 8
Let $\vec{w} = (1,1,1)$ Let $f: \mathbb{R}^3 \to \mathbb{R}^3$ be the endomorphism defined by:
$\vec{v} \in \mathbb{R}^3 \to f(\vec{v}) = (\vec{v} \cdot \vec{w})\vec{w} + \vec{v}$
Which one of the following statements is true?
(a) $\vec{w}$ is an eigenvector of eigenvalue 3
(b) f has no real eigenvalues.
(c) $\vec{w}$ is an eigenvector of eigenvalue 4
(d) If $\vec{v}$ is perpendicular to $\vec{w}$ then $\vec{v} \in \mathrm{Ker}(f)$
Precedente Successivo
Let Π be the plane R^3
$7x^2 + y^2 + 2x + y = 0$
$a_k = \begin{pmatrix} 4 & 3 & 1 \\ 8 & 6 & 0 \\ 1 & k & 3k-1 \end{pmatrix}$
$x - y - z = k + k^2$
$x + z = k.$
Let $\vec{w} = (1, 1, 2)$ Let $f: R^3$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1 2 3 4 5 6 7 8 9 10
Precedente Successivo
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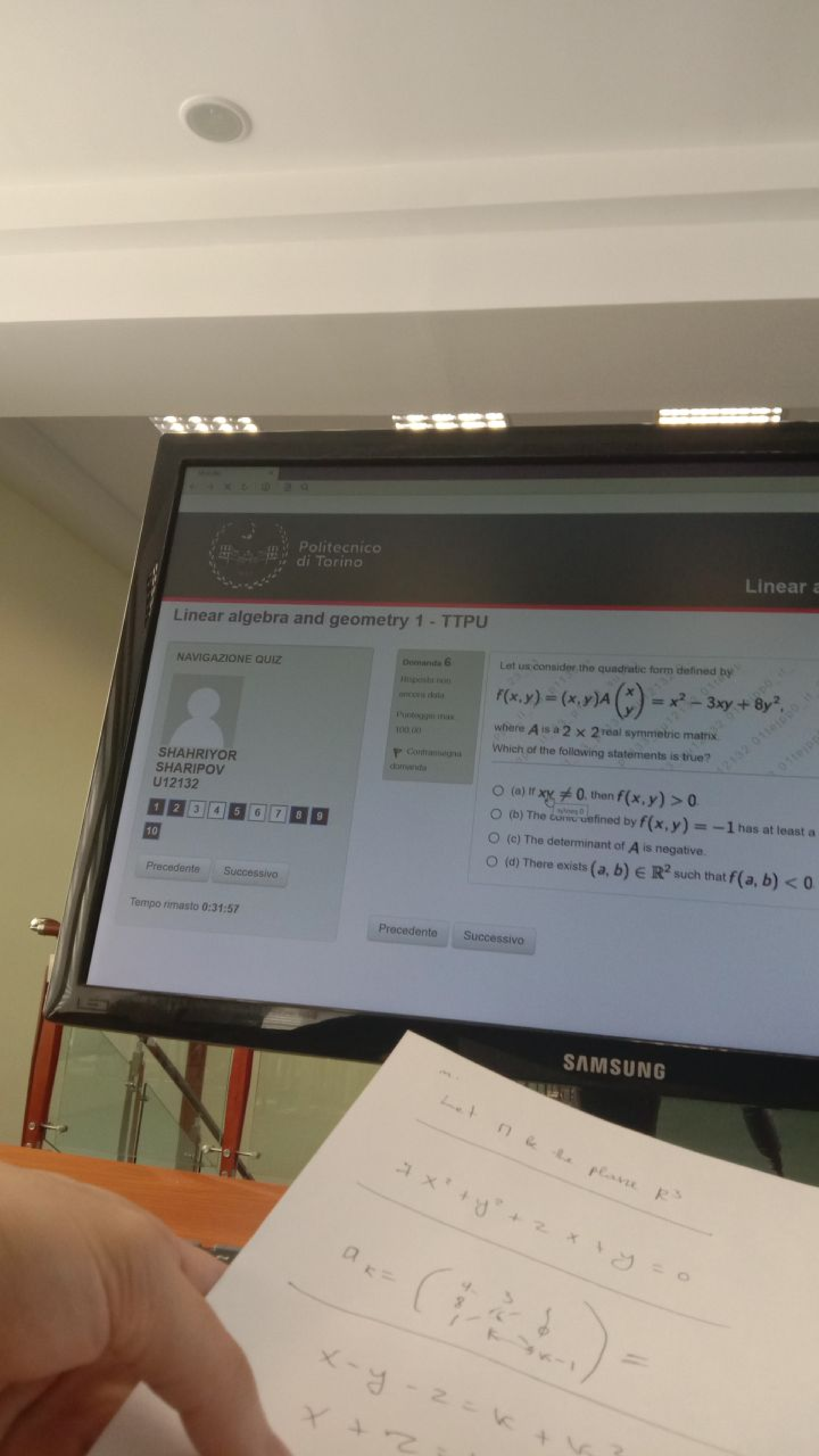
Domanda 5
Risposta non ancora data
Punteggio max. 100,00
Contrassegna domanda
Let $k \in \mathbb{R}$ be a real parameter and consider the matrix
$A_k = \begin{pmatrix} 8 & 6 & 0 \\ 4 & 3 & 1 \\ 1 & k & 3k-1 \end{pmatrix}$.
Find the correct statement.
(a) $\det(A_k) = 0$ for any $k \in \mathbb{R}$.
(b) A_k is invertible for any $k \neq \frac{3}{4}$.
(c) A_k is invertible for any $k \in \mathbb{R}$.
(d) $\det(A_k) \neq 0$ for any $k \in \mathbb{R}$.
Precedente Successivo
Let Π be the plane $\mathbb{R}^3$
$7x^2+y^2+2x+y=0$
$x-y-z=k+k^2$
$x+z=k$
$=(1,1,2)$ Let $f:\mathbb{R}^3$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Domanda 6
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Contrassegna domanda
Let us consider the quadratic form defined by
$f(x,y) = (x,y)A\begin{pmatrix} x \\ y \end{pmatrix} = x^2 - 3xy + 8y^2,$
where A is a 2×2 real symmetric matrix.
Which of the following statements is true?
(a) If $xy \neq 0$, then $f(x,y) > 0$.
(b) The conic defined by $f(x,y) = -1$ has at least a
(c) The determinant of A is negative.
(d) There exists $(a,b) \in \mathbb{R}^2$ such that $f(a,b) < 0$
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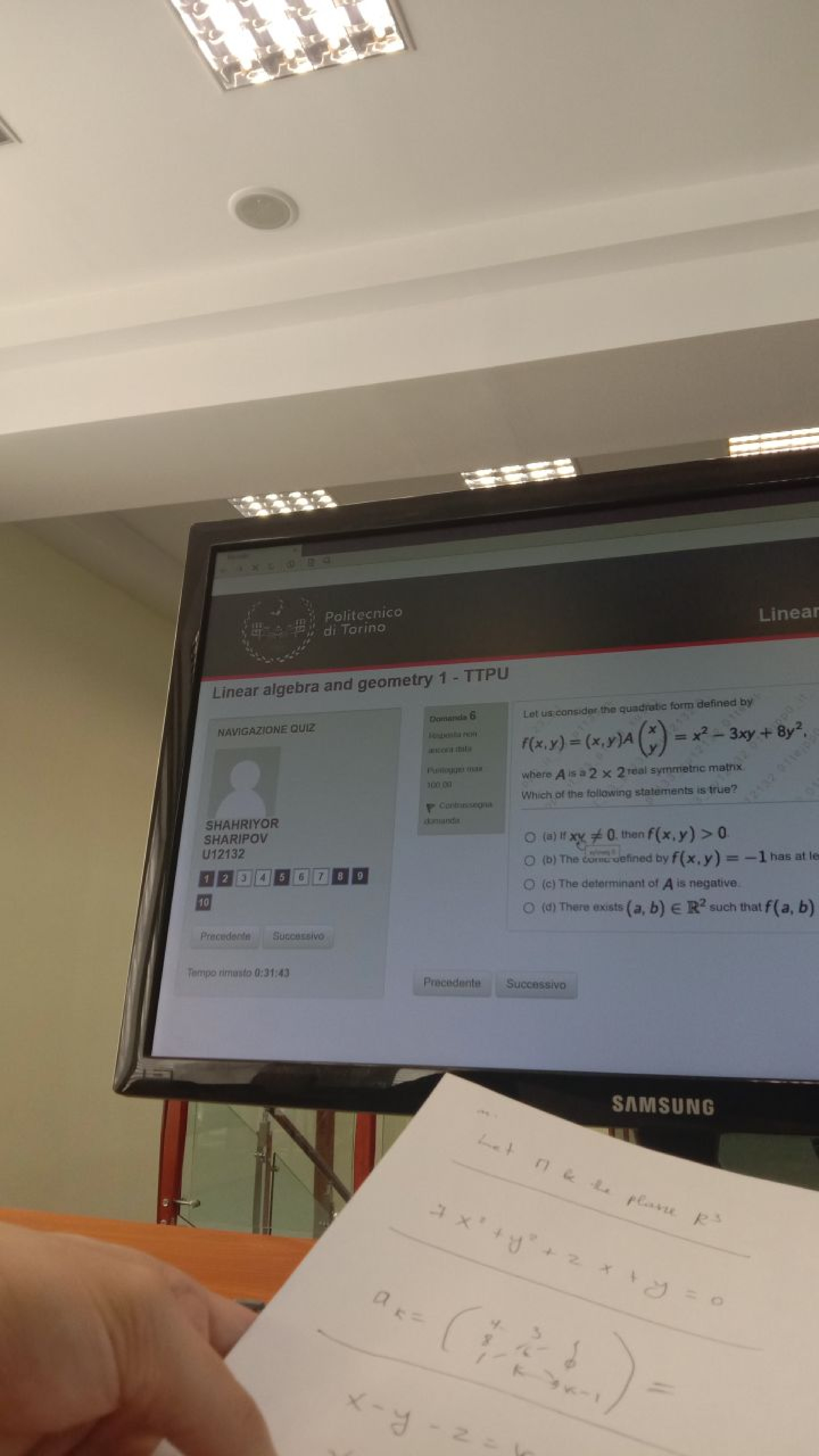
Let π be the plane $\mathbb{R}^3$
$7x^2 + y^2 + 2x + y = 0$
$a_k =$
$x - y - z =$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Domanda 6
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Contrassegna domanda
Let us consider the quadratic form defined by
$f(x,y) = (x,y)A\begin{pmatrix} x \\ y \end{pmatrix} = x^2 - 3xy + 8y^2,$
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(c) The determinant of A is negative.
(d) There exists $(a,b) \in \mathbb{R}^2$ such that $f(a,b)$
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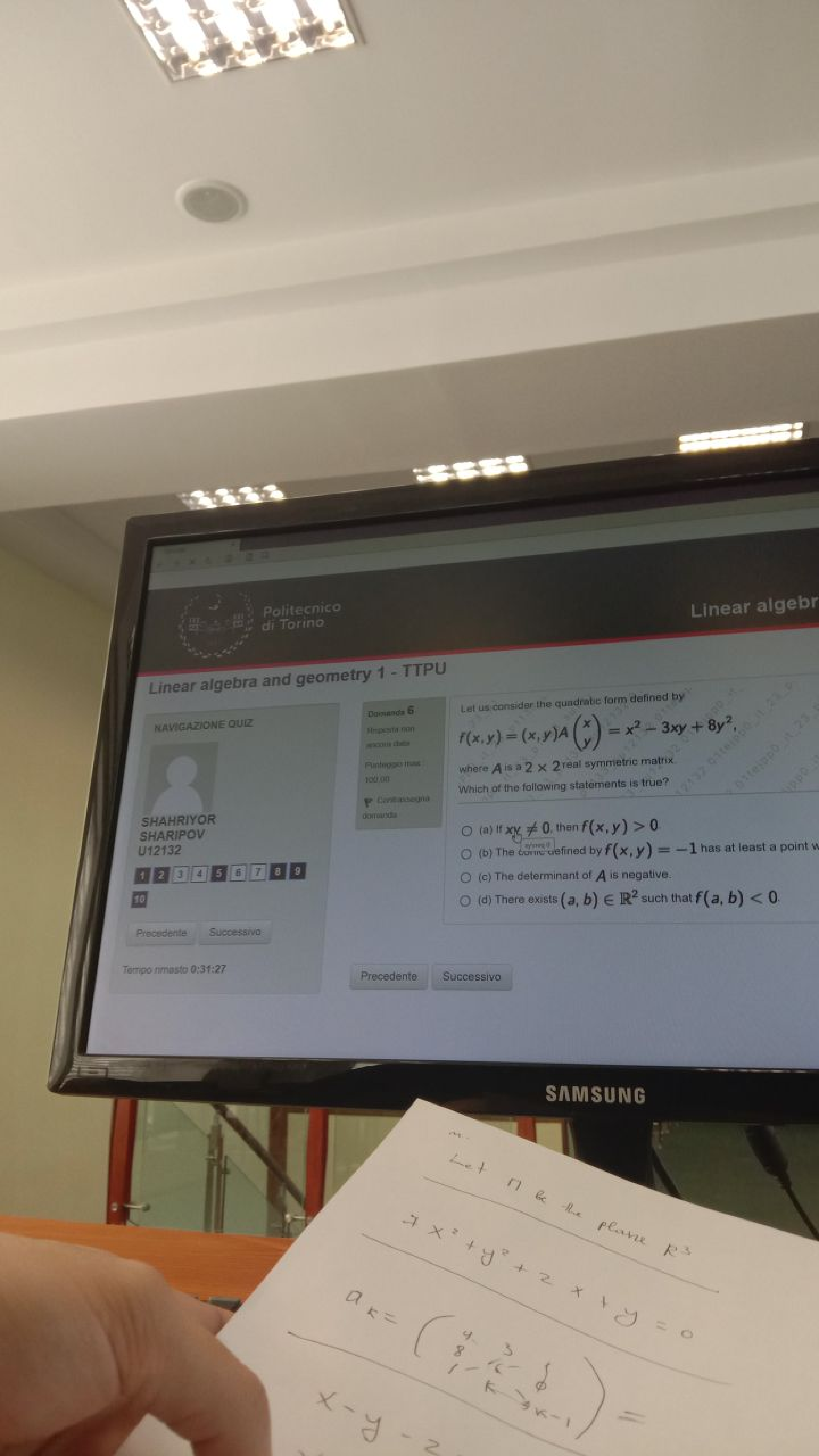
Let Π be the plane R^3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Let us consider the quadratic form defined by
$f(x, y) = (x, y)A\begin{pmatrix} x \\ y \end{pmatrix} = x^2 - 3xy + 8y^2$,
where A is a 2×2 real symmetric matrix.
Which of the following statements is true?
(a) If $xy \neq 0$, then $f(x, y) > 0$.
(b) The conic defined by $f(x, y) = -1$ has at least a point w
(c) The determinant of A is negative.
(d) There exists $(a, b) \in \mathbb{R}^2$ such that $f(a, b) < 0$.
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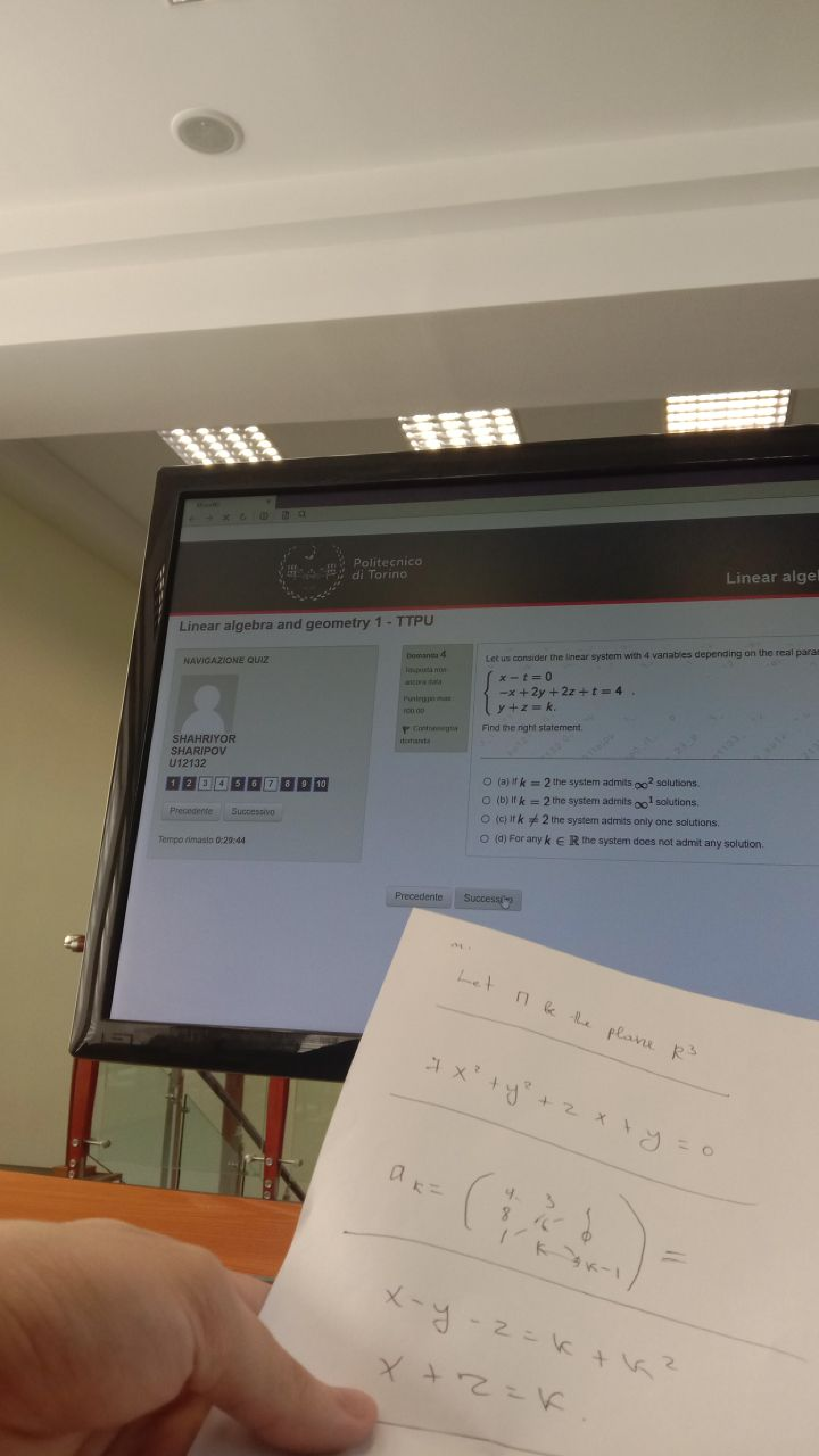
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Domanda 4
Let us consider the linear system with 4 variables depending on the real para
$\begin{cases} x - t = 0 \\ -x + 2y + 2z + t = 4 \\ y + z = k. \end{cases}$
Find the right statement.
(a) If $k = 2$ the system admits ∞^2 solutions.
(b) If $k = 2$ the system admits ∞^1 solutions.
(c) If $k \neq 2$ the system admits only one solutions.
(d) For any $k \in \mathbb{R}$ the system does not admit any solution.
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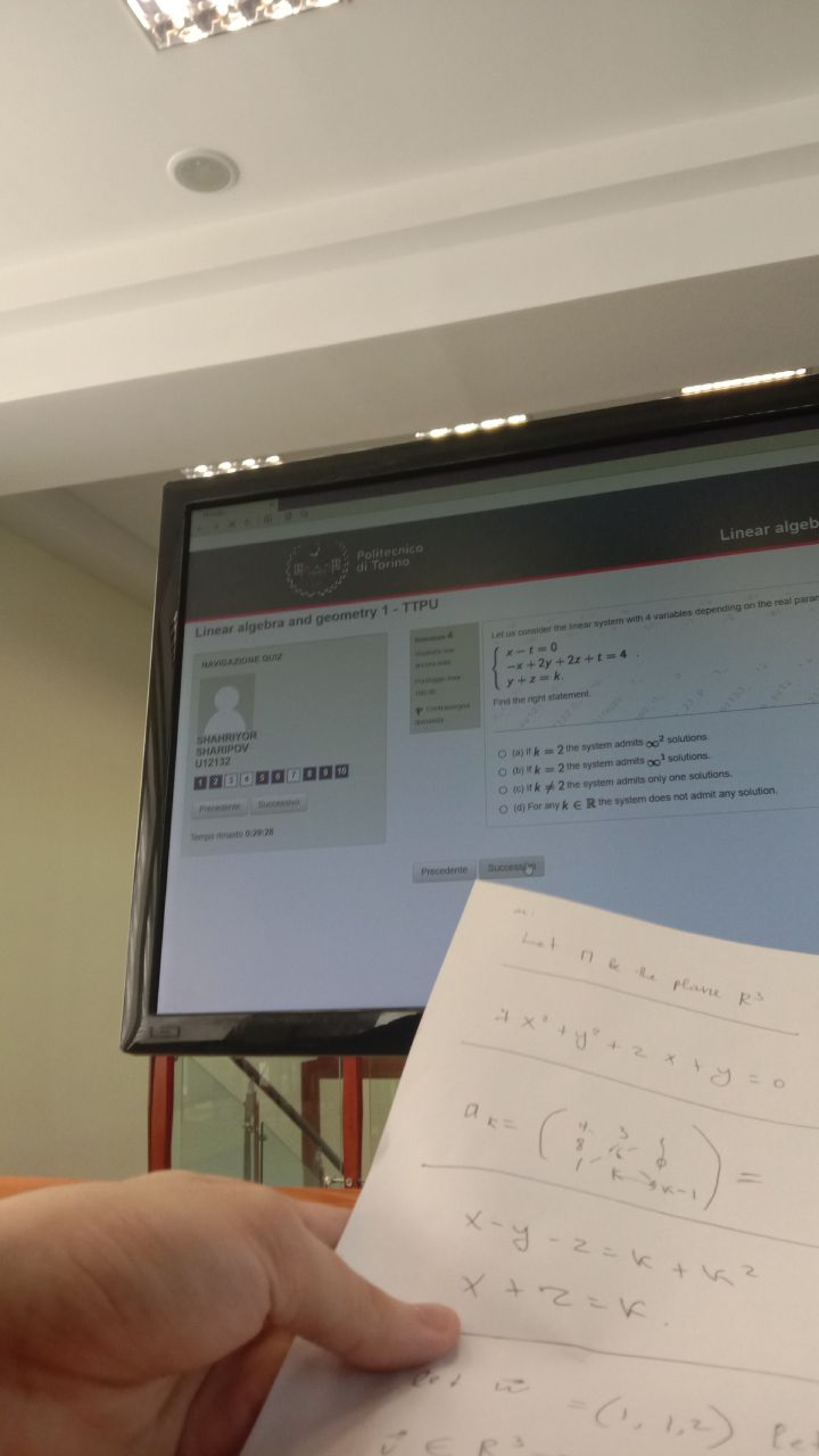
Let Π be the plane R^3
$7x^2 + y^2 + 2x + y = 0$
$x - y - z = k + k^2$
$x + z = k$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Let us consider the linear system with 4 variables depending on the real param
$x - t = 0$
$-x + 2y + 2z + t = 4$
$y + z = k.$
Find the right statement.
(a) If $k = 2$ the system admits ∞^2 solutions.
(b) If $k = 2$ the system admits ∞^1 solutions.
(c) If $k \neq 2$ the system admits only one solutions.
(d) For any $k \in \mathbb{R}$ the system does not admit any solution.
Precedente
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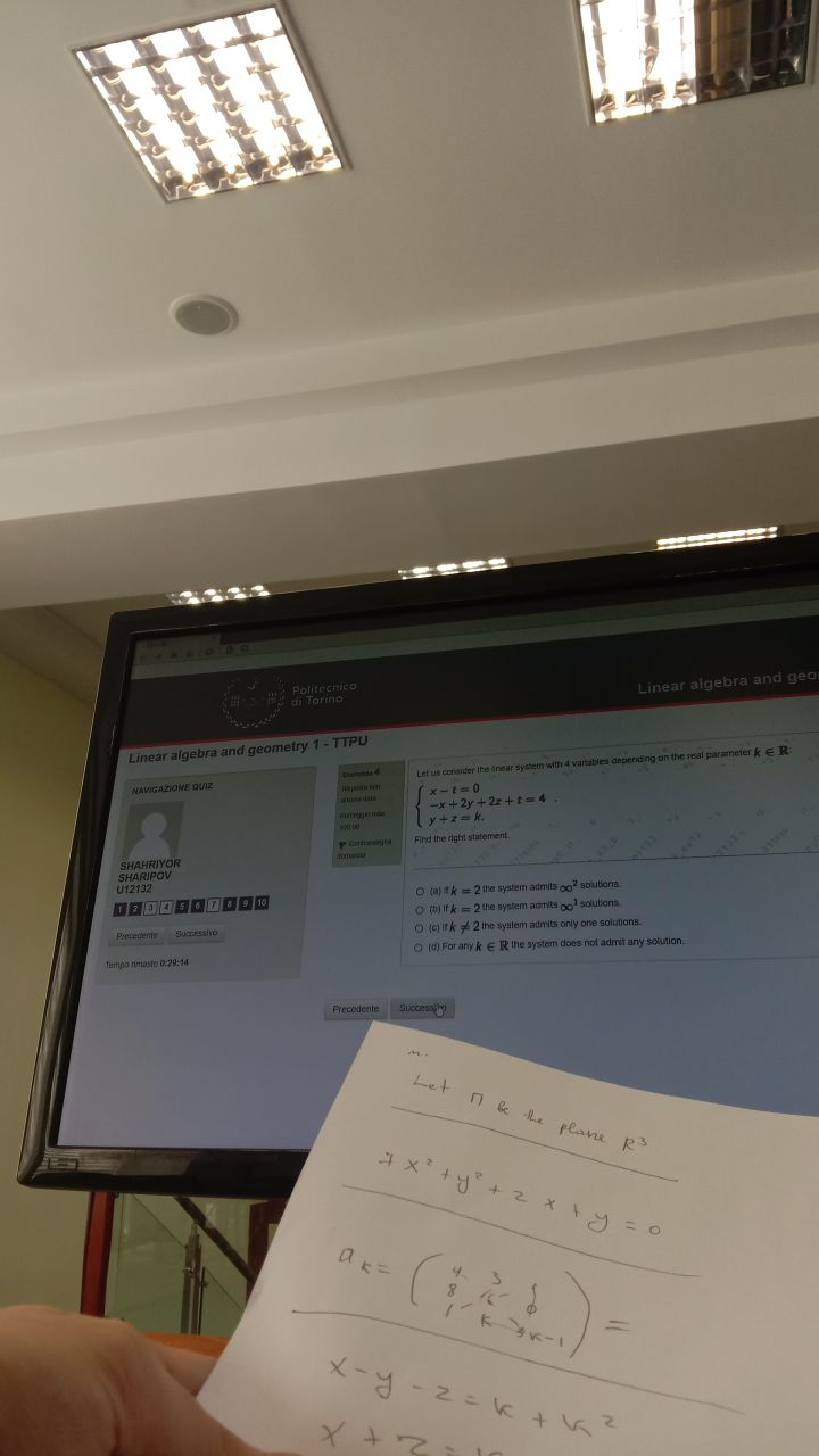
Let Π be the plane R³
4x² + y² + 2x + y = 0
x - y - z = k + k²
x + z = k.
= (1, 1, 2)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Domanda 4
Let us consider the linear system with 4 variables depending on the real parameter $k \in \mathbb{R}$:
$\begin{cases} x - t = 0 \\ -x + 2y + 2z + t = 4 \\ y + z = k. \end{cases}$
Find the right statement.
(a) If $k = 2$ the system admits ∞^2 solutions.
(b) If $k = 2$ the system admits ∞^1 solutions.
(c) If $k \neq 2$ the system admits only one solutions.
(d) For any $k \in \mathbb{R}$ the system does not admit any solution.
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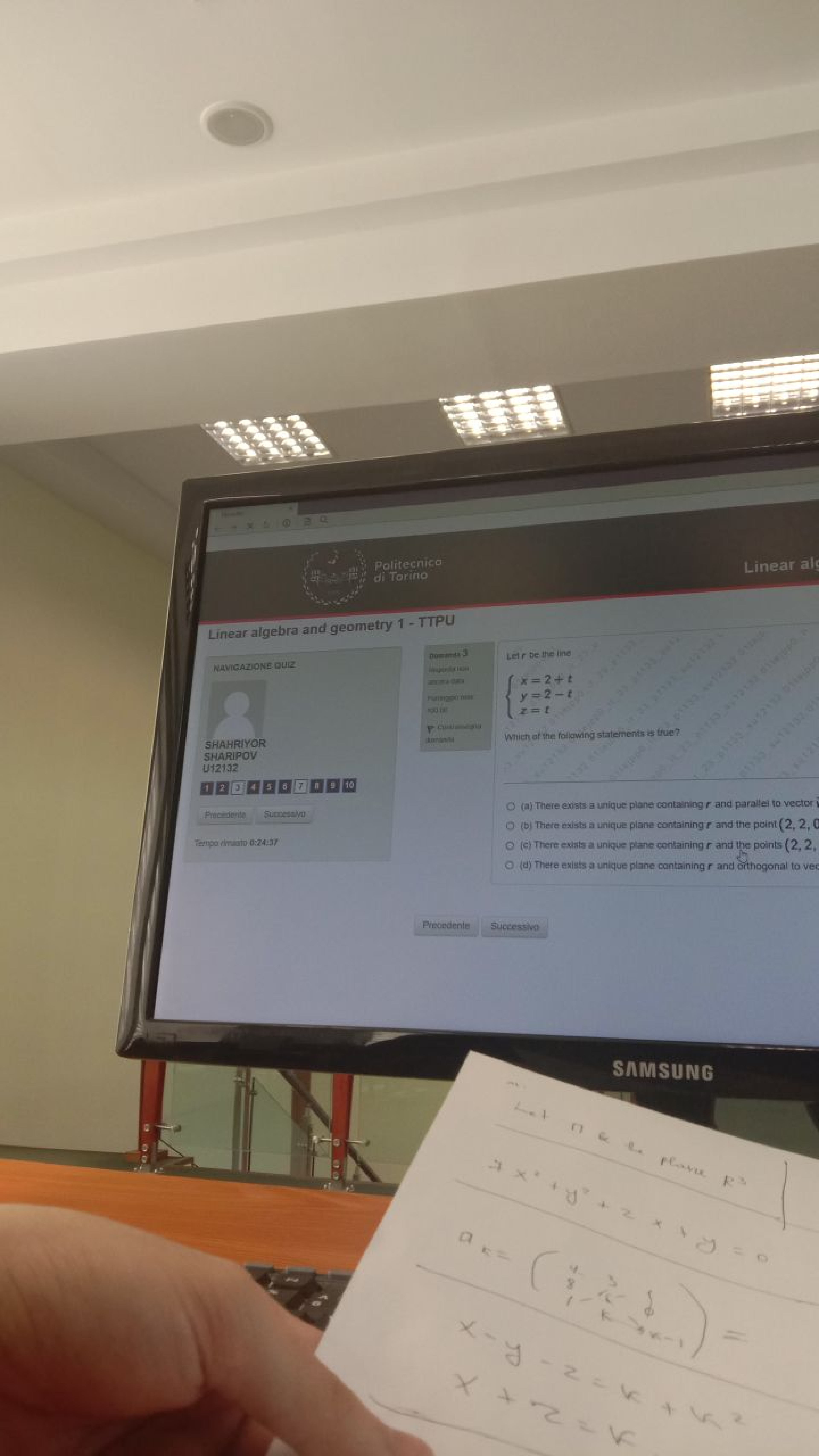
Let Π be the plane $\mathbb{R}^3$
$x^2 + y^2 + 2x + y = 0$
$x - y - z = k + k^2$



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Domanda 3
Let r be the line
x = 2 + t
y = 2 − t
z = t
Which of the following statements is true?
(a) There exists a unique plane containing r and parallel to vector
(b) There exists a unique plane containing r and the point (2, 2, 0
(c) There exists a unique plane containing r and the points (2, 2,
(d) There exists a unique plane containing r and orthogonal to vec
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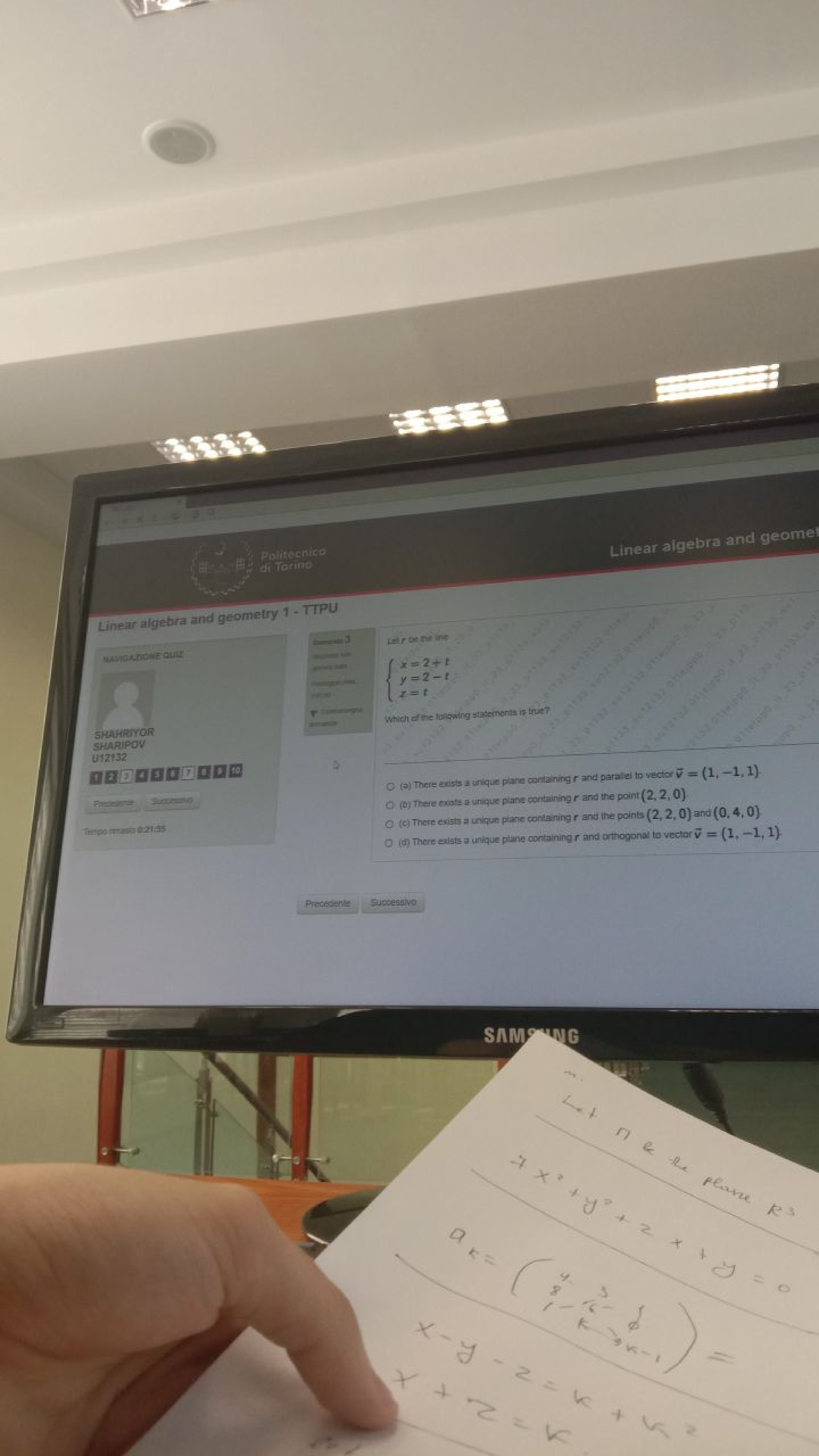
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Domanda 3
Let r be the line
$\begin{cases} x = 2 + t \\ y = 2 - t \\ z = t \end{cases}$
Which of the following statements is true?
(a) There exists a unique plane containing r and parallel to vector $\vec{v} = (1, -1, 1)$.
(b) There exists a unique plane containing r and the point $(2, 2, 0)$.
(c) There exists a unique plane containing r and the points $(2, 2, 0)$ and $(0, 4, 0)$.
(d) There exists a unique plane containing r and orthogonal to vector $\vec{v} = (1, -1, 1)$.
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Domanda 7

In the Euclidean plane with fixed cartesian system Oxy, let C be the conic defined by the equation
$7x^2 + y^2 + 2x + y = 0$.
Which of the following statements is correct?

- (a) C is a hyperbola.
- (b) C is degenerate.
- (c) C is an ellipse.
- (d) C is a parabola.

Precedente Successivo

m.

Let Π be the plane R^3

$x = 2 +$
$y = 2 - t$
$z = t$

$7x^2 + y^2 + 2x + y = 0$

$$a_k = \begin{pmatrix} 4 & 3 & 1 \\ 8 & 6 & \phi \\ 1 & k & k-1 \end{pmatrix} =$$

$x - y - z = k + k^2$

(5) $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ then A^{-1} $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

(6) Let us consider the quadratic form defined by $f(x,y) = (x,y) A \begin{pmatrix} x \\ y \end{pmatrix}$ with $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ True?

(a) there exits a pair $(a,b) \in R^2 \setminus \{(0,0)\}$ such that $q(a,b) = 0$

(b) $q(x,y)$ is positive definite

(c) The matrix A has two eigenvalues, ~~of~~ different sign

(d) $q(x,y) - 1 = 0$ is equation of a hyperbola

(7) Let $\vec{w} = (1,1,1)$. Let $f: R^3 \to R^3$ be the endomorphism defined by $\vec{v} \in R^3 \to f(\vec{v}) = (\vec{v} \cdot \vec{w})\vec{w} + 2\vec{v}$

(a) $\vec{w}$ isn't an eigenvector

(b) None of the above

√(c) If $\vec{v} \neq \vec{0}$ is perpendicular to $\vec{w}$ the $\vec{v}$ is an eigenvector

(d) f has no real eigenvalues

(8) Let us consider matrix $A = \begin{pmatrix} 2 & 0 & h \\ 0 & 3 & 0 \\ h & 0 & 2 \end{pmatrix}$ with $h \in R$

(a) A is diagonalizable with only $h = 0$

(b) A can be diagonalized for all $h \in R$

(c) A is diagonalizable only if $h \neq 2$

√(d) A has 3 distinct eigenvalues for all $h \in R$

(9) $x^2 + (-6xy) + 9y^2 + 2x - 6y = 0$ degenerate

(10) plane $\pi: 3x - 5y - z = 0$ and straight line r: $\begin{cases} 2x + y - z = 1 \\ 3x + 2y + z = 1 \end{cases}$ then

(a) $r \subseteq \pi$

√(b) none correct

(c) r and π orthogonal

(d) $r \cap \pi = \emptyset$

① if the linear system $Ax=b$ with 5 equations and 5 variables has ∞^m solutions with $m \le 3$ then necessarly

ⓐ rank $\left(\frac{A}{B}\right) = 2$

√ ⓑ rank $(A) \ge 2$

ⓒ rank $(A) = 2$

ⓓ rank $(A) \le 2$

② Let i, j, k the versors of the coordinate axes of R^3. For every $a \in R$ let us consider the vectors $\vec{u} = a\vec{j} + a\vec{k}$, $\vec{v} = 2\vec{i} + \vec{I}$, $\vec{w} = \vec{I}$

√ ⓐ There exists a unique $a \in R$ that vectors coplanar

ⓑ there exists a unique $a \in R$ that vectors linearly independent

ⓒ For every $a \in R$ vectors coplanar

ⓓ For every $a \in R$ vectors linearly independent

③ In the Euclidean 3-dimensional space with a fixed cartesian system, let us consider

$C_1: x^2 + y^2 + z^2 - 1 = x = 0$

$C_2: x^2 + y^2 + z^2 - 1 = x - z = 0$ having radius respectively

ⓐ $R_1 = R_2$

ⓑ $R_1 < R_2$

ⓒ C_1 consists of only one point

√ ⓓ $R_1 > R_2$

④ In R^4 let us consider the vector subspace

$V = \{(x, y, z, w) \mid x + y + z + w = 0,\ x - 3y = 0\}$ True?

√ ⓐ V is the vector subspace generated by the vectors $(3, 1, 0, -4)$ e $(0, 0, 1, -1)$

ⓑ A basis of V is formed by 3 vectors

ⓒ V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, -1)$

ⓓ V is subspace of dimension 1

20.01.24

iExam _ Linear Algebra and Geometry.

1. In the euclidean 3-dimensional space with fixed cartesian system.

$x^2+y^2+z^2-2y+2z=0$

$\boxed{x^2+y^2-2y+z^2+2z=0}$

$x^2+(y-1)^2-1+(z+1)^2-1=0$

$x^2+(y-1)^2+(z+1)^2=2$

$\boxed{R_1=0}$ $\boxed{R_2=1}$ $\boxed{R_3=-1}$

$\boxed{\text{The center } S \text{ has coordinates } (1,1,-1)}$

3. $A=[1\ 1\ 1]$ $B=[4\ 4]$

$A \cdot B = ?$

$A_{1\times3} \cdot B_{1\times2} = B \cdot A_{3\times2}$

$\Rightarrow B \cdot A = \begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \end{bmatrix}$

2. Let $V = \{(x,y,z) \in \mathbb{R}^3 ; x+y=0\}$ and $f: \mathbb{R}^3 \to \mathbb{R}^3$ be an endomorphism

$V = \ker(f)$ and 2 is an eigenvalue of f

- f is simple.
- $\dim(\mathrm{Im}(f)) = 2$
- f has three distinc eigenvalues.
- The characteristics polynomial of f can be $-t(t-2)^2$

4. $\mathbb{R}^{2,2}$ be the vectors space of real matrices with two rows and colums.

subsets of $\mathbb{R}^{2,2}$.

$V = \{\begin{pmatrix} 0 & a \\ b & c \end{pmatrix} \mid a,b,c \in \mathbb{R}\}$ and $W = \{\begin{pmatrix} d & e \\ e & 0 \end{pmatrix} \mid d,e \in \mathbb{R}\}$

- $\dim(V) = \dim(W) = 3$
- $\dim(V \cap W) = 1$
- $\dim(V \cap W) = 0$
- the Matrices $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}$ from basis of W

5. Let $\mathbb{R}^{3,2}$ be the vector space of real matrices with three rows and two colums.
and $\mathbb{R}^{2,2}$ be vector space of real matrices two rows and colums.
Let us consider linear map $f: \mathbb{R}^{3,2} \to \mathbb{R}^{2,2}$.

$f(X) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \cdot X \quad \forall X \in \mathbb{R}^{3,2}$

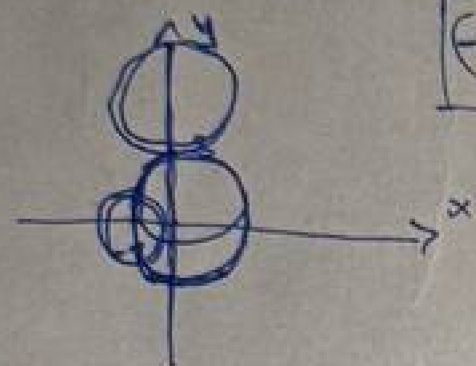
(6.) In the euclidean 2-dimensional of space

$x^2 + y^2 - 2x - 6y + 10 = 0$

$(x^2 - 2x + 1) - 1 + (y^2 - 6y + 9) - 9 + 10 = 0$

$(x+1)^2 + (y-3)^2 = 0$

$\boxed{(-1, 3) \,/\, R = 0}$



C is union of two non parallel lines.

(7.) $r: \begin{cases} x - y + 2z = 0 \\ 2x + y + z = 1 \end{cases}$ $s: \begin{cases} x + 2y - z = 1 \\ x + z = 1 \end{cases}$

Lines intersect, screw, parallel distinct, coincide $r = s$

(1) $\begin{cases} 2x - 2y + 4z = 0 \\ 2x + y + z = 1 \end{cases}$

$-3y + 3z = -1$

$-3(y - z) = -1$

$\boxed{y - z = \frac{1}{3}}$

(2) $2y - 2z = 0$

$\boxed{y - z = 0}$

what is screw

???

8. Euclidean 3-space

$A = (0, a, 0)$ $B = (3, a, -3)$ $C = (3, a, 3)$

- Area of ΔABC equal to 9 / $\frac{3a^2}{2}$ for every $a \in \mathbb{R}$

(9) Let S be system of 4 equations 3 variables. $M = (A|B)$ complete matrix associated to S.

det (M) = 0

- null vector is solution of S
- S admits only one solution / infinite / any type.

(10). $q(x,y)$ $A\begin{pmatrix} x \\ y \end{pmatrix} = -2x^2 + 4xy + y^2$.

A is 2X2 symmetric matrix

- All eigenvalues of A are negative.

HELP CENTER

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A) = 0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x, y, z) = (-x + z, -2y, x - z)$.

Which of the following statements is correct?

(a) All the eigenspaces of f have dimension 1.

(b) The image of f has dimension 1.

(c) f is simple.

(d) f is an isomorphism. ✗

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ \ldots \end{pmatrix}$$

HELP CENTER

(a) There exists a unique plane containing the point $(2,2,0)$ and orthogonal to r.

(b) There exists a unique plane containing the point $(2,2,0)$ and parallel to r.

(c) There exists a unique plane containing r and the point $(2,2,0)$. ✗

(d) There exists a unique plane containing r and orthogonal to the vector $\vec{v}=(1,-1,1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2,2,0)$ and orthogonal to r.

Question 6

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zer
and find the correct statement.

(a) $\det(A)=0$.

(b) $\operatorname{rank}(A)=n$. ✗

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A)=0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x,y,z)=(-x+z,-2y,x-z)$.

Which of the following statements is correct?

100.00

Flag question

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(d) C_1 and C_2 coincide. ✗

Risposta errata.

The correct answer is: $R_2 > R_1$.

Question 5

Incorrect

Mark -15.00 out of 100.00

Flag question

Let us consider the line

$$r: \begin{cases} x = 2 + t \\ y = 2 - t \\ z = t \end{cases}$$

Which of the following statements is true?

(a) There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

(b) There exists a unique plane containing the point $(2, 2, 0)$ and parallel to r.

(c) There exists a unique plane containing r and the point $(2, 2, 0)$.
✗

(d) There exists a unique plane containing r and orthogonal to the vector $\vec{v} = (1, -1, 1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

HELP CENTER

Question 3

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

(c) 0 is not an eigenvalue of A.

(d) None of the other statements is correct.

Risposta errata.

The correct answer is: A is not positive or negative definite.

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lag question

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1 : x + y + z = 2$ $\pi_2 : x + y + z = 0$ let us denote by R the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(d) C_1 and C_2 coincide. ✗

Risposta errata.

The correct answer is: $R_2 > R_1$.

n 5

Let us consider the line

HELP CENTER

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t-3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

Question 3

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

(c) 0 is not an eigenvalue of A.

(d) None of the other statements is correct.

Risposta errata.

The correct answer is: A is not positive or negative definite.

stion 4

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-15.00 out of 0

ag question

Given the sphere $S: x^2+y^2+z^2-2x+2y=7$, and the planes $\pi_1: x+y+z=2$ $\pi_2: x+y+z=0$ let us den

the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(c) $\dim(\text{Ker } f) = 2$.

(d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\text{Ker } f) = 1$.

Question 2

Incorrect

Mark -15.00 out of 100.00

Flag question

For all $t \in \mathbb{R}$, consider the points

$$P = (1, 2, 3), \quad Q = (0, 0, t), \quad R = (1, -1, t), \quad S = (1, 1, t).$$

Which of the following statements is true?

(a) Points P, Q, R, S are vertices of a parallelogram for all values of t.

(b) Points P, Q, R, S are coplanar for a unique value of t.

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

estion 3

orrect

rk -15.00 out of

.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

State	Finished
Completed on	Tuesday, 18 January 2022, 9:02 AM
Time taken	1 hour
Marks	-35.00/1000.00
Grade	**-1.16** out of 33.00 (**-4**%)

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-15.00 out of
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lag question

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f : \mathbb{R}_2[x] \to \mathbb{R}^3$ be a linear map such that

$$f(x^2) = (0, 0, 1), \quad f(x) = (0, 1, 0), \quad f(1) = (0, 3, 3).$$

Which one of the following statements is true?

- (a) $f(3x - 1) = (0, 0, 3)$. ✗
- (b) $\dim(\operatorname{Ker} f) = 1$.
- (c) $\dim(\operatorname{Ker} f) = 2$.
- (d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\operatorname{Ker} f) = 1$.

tion 2

rect

For all $t \in \mathbb{R}$, consider the points

$P = (1, 2, 3)$, $Q = (0, 0, t)$, $R = (1, -1, t)$, $S = (1, 1, t)$

HELP CENTER

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $\mathbb{R}_3[x]$ is a vector subspace?

- (a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.
- (b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗
- (c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.
- (d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let $\mathcal{C}$ be the conic defined by the equation

$$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0.$$

Which of the following statements is correct?

- (a) $\mathcal{C}$ is a hyperbola.
- (b) $\mathcal{C}$ is degenerate. ✗
- (c) $\mathcal{C}$ is a parabola.
- (d) $\mathcal{C}$ is an ellipse.

Risposta errata.

The correct answer is: $\mathcal{C}$ is a parabola.

HELP CENTER

(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$ ✓

Risposta corretta.

The correct answer is: $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $\mathbb{R}_3[x]$ is a vector subspace?

- (a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.
- (b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗
- (c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.
- (d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let C be the conic defined by the equation

$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0.$

Which of the following statements is correct?

- (a) C is a hyperbola.

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Flag question

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & -1 & 2 \end{pmatrix}$$

Find the correct statement.

(a) $A^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & -1 & -3 \\ 1 & -1 & -1 \end{pmatrix}$

(b) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(c) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 2 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$ ✓

HELP CENTER

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A) = 0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x, y, z) = (-x + z, -2y, x - z)$.

Which of the following statements is correct?

(a) All the eigenspaces of f have dimension 1.

(b) The image of f has dimension 1.

(c) f is simple.

(d) f is an isomorphism.

✗

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ \end{pmatrix}$$

HELP CENTER

(a) There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

(b) There exists a unique plane containing the point $(2, 2, 0)$ and parallel to r.

(c) There exists a unique plane containing r and the point $(2, 2, 0)$.

(d) There exists a unique plane containing r and orthogonal to the vector $\vec{v} = (1, -1, 1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

Question 6

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zer
and find the correct statement.

(a) $\det(A) = 0$.

(b) $\text{rank}(A) = n$.

(c) A is an invertible matrix.

(d) None of the previous statements is correct.

Risposta errata.

The correct answer is: $\det(A) = 0$.

Question 7

Incorrect

Mark -15.00 out of 100.00

We recall that an endomorphism $f : V \to V$ is said to be simple if there exists a basis of V of eigenvectors of f.

Let us consider the endomorphism $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$f(x, y, z) = (-x + z, -2y, x - z)$.

Which of the following statements is correct?

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Flag question

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(d) C_1 and C_2 coincide. ✗

Risposta errata.

The correct answer is: $R_2 > R_1$.

Question 5

Incorrect

Mark -15.00 out of 100.00

Flag question

Let us consider the line

$$r: \begin{cases} x = 2 + t \\ y = 2 - t \\ z = t \end{cases}$$

Which of the following statements is true?

(a) There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

(b) There exists a unique plane containing the point $(2, 2, 0)$ and parallel to r.

(c) There exists a unique plane containing r and the point $(2, 2, 0)$.

✗

(d) There exists a unique plane containing r and orthogonal to the vector $\vec{v} = (1, -1, 1)$.

Risposta errata.

The correct answer is: There exists a unique plane containing the point $(2, 2, 0)$ and orthogonal to r.

HELP CENTER

Question 3
Incorrect
Mark -15.00 out of 100.00
Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.
Find the right statement.

(a) A is not positive or negative definite.
(b) A is positive or negative semi-definite. ✗
(c) 0 is not an eigenvalue of A.
(d) None of the other statements is correct.

Risposta errata.
The correct answer is: A is not positive or negative definite.

estion 4
orrect
k -15.00 out of 00
lag question

Given the sphere $S: x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1: x + y + z = 2$ $\pi_2: x + y + z = 0$ let us denote by R the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.
(b) $R_2 = R_1$.
(c) $R_2 > R_1$.
(d) C_1 and C_2 coincide. ✗

Risposta errata.
The correct answer is: $R_2 > R_1$.

n 5
Let us consider the line

HELP CENTER

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t-3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

Question 3

Incorrect

Mark -15.00 out of 100.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

(c) 0 is not an eigenvalue of A.

(d) None of the other statements is correct.

Risposta errata.

The correct answer is: A is not positive or negative definite.

stion 4

rect

-15.00 out of 0

ag question

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1 : x + y + z = 2$ $\pi_2 : x + y + z = 0$ let us den

the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_1 > R_2$.

(b) $R_2 = R_1$.

(c) $R_2 > R_1$.

(c) $\dim(\operatorname{Ker} f) = 2$.

(d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\operatorname{Ker} f) = 1$.

Question 2

Incorrect

Mark -15.00 out of 100.00

Flag question

For all $t \in \mathbb{R}$, consider the points

$$P = (1, 2, 3), \quad Q = (0, 0, t), \quad R = (1, -1, t), \quad S = (1, 1, t).$$

Which of the following statements is true?

(a) Points P, Q, R, S are vertices of a parallelogram for all values of t.

(b) Points P, Q, R, S are coplanar for a unique value of t.

(c) Points P, Q, R, S are vertices of a rectangle for all values of t.

(d) Points P, Q, R, S are vertices of a tetrahedron whose volume is $|t - 3|$ for all values of t. ✗

Risposta errata.

The correct answer is: Points P, Q, R, S are coplanar for a unique value of t.

estion 3

orrect

rk -15.00 out of

.00

Flag question

Let A be a real 6×6 non-null symmetric matrix with zero trace and determinant.

Find the right statement.

(a) A is not positive or negative definite.

(b) A is positive or negative semi-definite. ✗

State	Finished
Completed on	Tuesday, 18 January 2022, 9:02 AM
Time taken	1 hour
Marks	-35.00/1000.00
Grade	**-1.16** out of 33.00 (**-4**%)

stion **1**

rect

: -15.00 out of

00

lag question

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f: \mathbb{R}_2[x] \to \mathbb{R}^3$ be a linear map such that

$$f(x^2) = (0, 0, 1), \quad f(x) = (0, 1, 0), \quad f(1) = (0, 3, 3).$$

Which one of the following statements is true?

- (a) $f(3x - 1) = (0, 0, 3)$. ✗
- (b) $\dim(\operatorname{Ker} f) = 1$.
- (c) $\dim(\operatorname{Ker} f) = 2$.
- (d) f is surjective.

Risposta errata.

The correct answer is: $\dim(\operatorname{Ker} f) = 1$.

tion **2**

rect

For all $t \in \mathbb{R}$, consider the points

$P = (1, 2, 3)$, $Q = (0, 0, t)$, $R = (1, -1, t)$, $S = (1, 1, t)$

HELP CENTER

Politecnico di Torino

Linear algebra and geometry 1 - TTPU

and geometry 1 - TTPU

7 8 9 10

Iniziato	venerdì, 4 febbraio 2022, 08:04
Stato	Completato
Terminato	venerdì, 4 febbraio 2022, 09:04
Tempo impiegato	1 ora
Punteggio	655,00/1000,00
Valutazione	**21,62** su un massimo di 33,00 (**66**%)

Domanda **1**

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = x + y = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2 - 1 = x = 0,$$

having radius respectively R_1 and R_2.

Which of the following statements is true?

- (a) $R_1 > R_2$.
- (b) $\mathcal{C}_1$ consists of only one point.
- (c) $R_1 < R_2$. ✗
- (d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$

Domanda **2**

In the euclidean space let us consider the points

HELP CENTER

(b) C_1 consists of only one point.

(c) $R_1 < R_2$. ✗

(d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$.

Domanda 2

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

In the euclidean space let us consider the points

$A = (0, 1 + a, 1)$, $B = (0, 2 + a, 1)$, $C = (2, 1, 0)$, $D = (0, 1, 0)$.

Which of the following statements is true?

(a) There exists only one $a \in \mathbb{R}$ such that the points are collinear.

(b) For every $a \in \mathbb{R}$ the points are not coplanar. ✓

(c) For every $a \in \mathbb{R}$ the points are collinear.

(d) There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

Risposta corretta.

La risposta corretta è: For every $a \in \mathbb{R}$ the points are not coplanar.

Domanda 3

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let U and V be vector subspaces of $\mathbb{R}^5$ such that $dim(U) = 3$ e $dim(V) = 4$.

Which one of the following statements is always true?

(a) $U + V = \mathbb{R}^5$.

(b) $dim(U \cap V) \geq 2$. ✓

(d) There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

Risposta corretta.

La risposta corretta è: For every $a \in \mathbb{R}$ the points are not coplanar.

Domanda 3

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let U and V be vector subspaces of $\mathbb{R}^5$ such that $dim(U) = 3$ e $dim(V) = 4$.

Which one of the following statements is always true?

- (a) $U + V = \mathbb{R}^5$.
- (b) $dim(U \cap V) \geq 2$. ✓
- (c) $dim(U \cap V) = 1$.
- (d) $U \subseteq V$.

Risposta corretta.

La risposta corretta è: $dim(U \cap V) \geq 2$

Domanda 4

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the quadratic form defined by

$$q(x, y) = (x, y)A\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

- (a) The quadratic form $q(x, y)$ is negative definite.

HELP CENTER

liano (it)

Risposta corretta.

La risposta corretta è: $\mathcal{C}$ is a parabola.

Domanda 9

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$.

Which one of the following statements is true?

- (a) f has two distinct eigenvalues.
- (b) $(2, 0)$ is an eigenvector. ✓
- (c) $(0, 1)$ is an eigenvector.
- (d) f is simple.

Risposta corretta.

La risposta corretta è: $(2, 0)$ is an eigenvector.

Domanda 10

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Given the straight lines

$$r : \begin{cases} 2x - 5y = 7 \\ 2z - 3y = 5 \end{cases} \quad , \quad \text{and} \quad s : \begin{cases} x - y - z = 1 \\ x - 4y + z = 6, \end{cases}$$

find the right statement.

- (a) The lines are skew. ✗
- (b) The lines intersect at exactly one point.
- (c) None of the other statements is correct.
- (d) The lines are parallel and distinct.

Risposta errata.

La risposta corretta è: None of the other statements is correct.

HELP CENTER

aliano (it)

La risposta corretta è: $\det(A \cdot B) = 5$.

Domanda 7

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

If the linear system $AX = B$ with 4 equations and 5 variables has ∞^m solutions with $m \geq 1$, then necessarily

(a) $\mathrm{rank}(A|B) = 4$.

(b) $\mathrm{rank}(A) = 4$.

(c) $\mathrm{rank}(A) \geq 4$.

(d) $\mathrm{rank}(A) \leq 4$. ✓

Risposta corretta.

La risposta corretta è: $\mathrm{rank}(A) \leq 4$.

Domanda 8

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic $\mathcal{C}$ having equation

$x^2 - 6xy + 9y^2 + 2x - 4y = 0$.

Which of the following statements is true?

(a) $\mathcal{C}$ is a hyperbola.

(b) $\mathcal{C}$ is an ellipse.

(c) $\mathcal{C}$ is a parabola. ✓

(d) $\mathcal{C}$ is degenerate.

Risposta corretta.

La risposta corretta è: $\mathcal{C}$ is a parabola.

Domanda 9

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$

HELP CENTER

italiano (it)

La risposta corretta è: $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

Domanda 5
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

Let $f : \mathbb{R}^4 \to \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - 3)$.

Which one of the following statements is true?

- (a) $dim(\text{Im}(f)) = 4$.
- (b) $dim(\text{Im}(f)) = 3$.
- (c) $dim(\text{Ker}(f)) = 2$. ✗
- (d) $dim(\text{Im}(f)) = 2$.

Risposta errata.
La risposta corretta è: $dim(\text{Im}(f)) = 4$.

Domanda 6
Risposta corretta
Punteggio ottenuto 100,00 su 100,00
Contrassegna domanda

Let $\cdot$ denote the usual matrix multiplication. If

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix},$$

then

- (a) $\det(\mathbf{A} \cdot \mathbf{B}) = 0$.
- (b) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$.
- (c) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$.
- (d) $\det(\mathbf{A} \cdot \mathbf{B}) = 5$. ✓

Risposta corretta.

HELP CENTER

ano (it)

Domanda 4

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the quadratic form defined by

$$q(x, y) = (x, y)A\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

- (a) The quadratic form $q(x, y)$ is negative definite.
- (b) $q(a, b) > 0$ for every $(a, b) \in \mathbb{R}^2 \setminus \{(0,0)\}$. ✓
- (c) The matrix A has two eigenvalues of different sign.
- (d) $q(x, y) = 0$ is the equation of two lines passing through the origin.

Risposta corretta.

La risposta corretta è: $q(a, b) > 0$ for every $(a, b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

Domanda 5

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Let $f : \mathbb{R}^4 \to \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - 3)$.

Which one of the following statements is true?

- (a) $dim(\text{Im}(f)) = 4$.
- (b) $dim(\text{Im}(f)) = 3$.
- (c) $dim(\text{Ker}(f)) = 2$. ✗
- (d) $dim(\text{Im}(f)) = 2$.

Risposta errata.

La risposta corretta è: $dim(\text{Im}(f)) = 4$.

Politecnico
di Torino

Linear a

and geometry 1 - TTPU

7 8 9 10

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Valutazione	**21,62** su un massimo di 33,00 (**66**%)

Domanda 1

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

In the euclidean 3-dimensional space with a fixed cartesian system, let us

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = x + y = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2$$

having radius respectively R_1 and R_2.

Which of the following statements is true?

- (a) $R_1 > R_2$.
- (b) $\mathcal{C}_1$ consists of only one point.
- (c) $R_1 < R_2$. ✗
- (d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$.

Domanda 2

In the euclidean space let us consider the points

HELP CENTER

Question 1

Not answered

Marked out of 100.00

Flag question

Let P, L, and U be the factors of the $PA = LU$ factorization of a given matrix A, and let $s = 2$ be the total number of performed row swaps. If

$$U = \begin{pmatrix} 2 & 5 & 7 \\ 0 & 1 & 9 \\ 0 & 0 & 3 \end{pmatrix}$$

then the determinant of A is equal to:

- 2
- 6
- -6
- -2

The correct answer is: 6

Question 2

Correct

Mark 100.00 out of 100.00

Flag question

Consider a floating point arithmetic with $N = 10$, $t = 3$ digit reserved for the mantissa and rounding technique "rounding to even". Let p be the mantissa and $\bar{p}$ the corresponding machine mantissa. If $p = 0.13819$, then:

(a) $\bar{p} = 0.1382$

(b) $\bar{p} = 0.138$ ✓

(c) $\bar{p} = 0.139$

(d) $\bar{p} = 0.1381$

HELP CENTER

Question 2

Correct

Mark 100.00 out of 100.00

Flag question

Consider a floating point arithmetic with $N = 10$, $t = 3$ digit reserved for the mantissa and rounding technique "rounding to even". Let p be the mantissa and $\bar{p}$ the corresponding machine mantissa. If $p = 0.13819$, then:

- (a) $\bar{p} = 0.1382$
- (b) $\bar{p} = 0.138$ ✓
- (c) $\bar{p} = 0.139$
- (d) $\bar{p} = 0.1381$

The correct answer is: $\bar{p} = 0.138$

Question 3

Not answered

Marked out of 100.00

Flag question

Given a matrix A of dimension $n \times n$, assume one applies to A the inverse power method, with the aim of finding the eigenvalue λ closest to $p = 2$ and the corresponding eigenvector x. Let wk be a MATLAB variable containing the approximation of x at the k-th iteration of the method.

Which one of the following MATLAB instructions creates a variable lambdak containing the approximation of λ at step k?

- zk = (A - 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);
- zk = (A + 2*eye(n))\wk; lambdak = 2 - 1/(wk'*zk);
- zk = (A + 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);
- zk = (A - 2*eye(n))\wk; lambdak = 2 - 1/(wk'*zk);

The correct answer is: zk = (A - 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);

HELP CENTER

Mark -15.00 out of 100.00

Flag question

- A(:,end)
- col(10,A) ✗
- col(A,10)
- A('all',10)

The correct answer is: A(:,end)

Question 5

Not answered

Marked out of 100.00

Flag question

Let the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 4 & 5 \end{pmatrix}$$

be given, and assume to apply the power method to A. Let $w^{(m)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ be the approximation of the eigenvector corresponding to the eigenvalue of maximum modulus λ_1 at step m.

What is the value of $\lambda_1^{(m)}$, that is the approximation of λ_1 at step m?

- 15
- 9
- 3
- 8

HELP CENTER

The correct answer is: 3

Question 6

Not answered

Marked out of 100.00

Flag question

We are given a matrix $A \in R^{n,n}$, symmetric and positive definite, and its Choleski factorization $A = R^T R$. Which one of the following statements is surely correct?

- the factorization always exists but it is not unique
- the uniqueness of the factorization depends od the rank of matrix A
- the factorization does not always exist, but when it exists it is unique
- the factorization always exists and it is unique

The correct answer is: the factorization always exists and it is unique

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

Consider the following set of Matlab commands:

```
x=linspace(-1,1,10);
xx=linspace(-1,1);
y=sin(pi*x);
z=spline(x,y,xx);
```

What is their result?

- They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$ ✗
- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$

HELP CENTER

The correct answer is: the factorization always exists and it is unique

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

Consider the following set of Matlab commands:

```
x=linspace(-1,1,10);
xx=linspace(-1,1);
y=sin(pi*x);
z=spline(x,y,xx);
```

What is their result?

- They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$. ✗
- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

The correct answer is: They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$

Question 8

Correct

Mark 100.00 out of 100.00

In exact arithmetic (base 10), the result of the multiplication

$513.21 * 20.135$

represented in the normalized floating point notation is:

HELP CENTER

spaced points in $[-1, 1]$.

- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

The correct answer is: They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

Question 8

Correct

Mark 100.00 out of 100.00

Flag question

In exact arithmetic (base 10), the result of the multiplication

$513.21 * 20.135$

represented in the normalized floating point notation is:

- $10.33348337 * 10^3$
- $0.10333 * 10^6$
- $0.1033348335 * 10^5$ ✓
- $1.0333 * 10^4$

The correct answer is: $0.1033348335 * 10^5$

Finish review

HELP CENTER

Politecnico di Torino

Linear algebra and geometry 1 - TTPU

and geometry 1 - TTPU

7 8 9 10

Iniziato	venerdì, 4 febbraio 2022, 08:04
Stato	Completato
Terminato	venerdì, 4 febbraio 2022, 09:04
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Valutazione	**21,62** su un massimo di 33,00 (**66%**)

Domanda 1
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = x + y = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2 - 1 = x = 0,$$

having radius respectively R_1 and R_2.

Which of the following statements is true?

- (a) $R_1 > R_2$.
- (b) $\mathcal{C}_1$ consists of only one point.
- (c) $R_1 < R_2$. ✗
- (d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$

Domanda 2

In the euclidean space let us consider the points

HELP CENTER

(b) C_1 consists of only one point.

(c) $R_1 < R_2$.

(d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$.

Domanda 2

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

In the euclidean space let us consider the points

$$A = (0, 1 + a, 1), \quad B = (0, 2 + a, 1), \quad C = (2, 1, 0), \quad D = (0, 1, 0).$$

Which of the following statements is true?

(a) There exists only one $a \in \mathbb{R}$ such that the points are collinear.

(b) For every $a \in \mathbb{R}$ the points are not coplanar.

(c) For every $a \in \mathbb{R}$ the points are collinear.

(d) There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

Risposta corretta.

La risposta corretta è: For every $a \in \mathbb{R}$ the points are not coplanar.

Domanda 3

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let U and V be vector subspaces of $\mathbb{R}^5$ such that $dim(U) = 3$ e $dim(V) = 4$.

Which one of the following statements is always true?

(a) $U + V = \mathbb{R}^5$.

(b) $dim(U \cap V) \geq 2$.

HELP CENTER

(d) There exists only one $a \in \mathbb{R}$ such that the points are not coplanar.

Risposta corretta.

La risposta corretta è: For every $a \in \mathbb{R}$ the points are not coplanar.

Domanda 3

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let U and V be vector subspaces of $\mathbb{R}^5$ such that $dim(U) = 3$ e $dim(V) = 4$.

Which one of the following statements is always true?

(a) $U + V = \mathbb{R}^5$.

(b) $dim(U \cap V) \geq 2$.

(c) $dim(U \cap V) = 1$.

(d) $U \subseteq V$.

Risposta corretta.

La risposta corretta è: $dim(U \cap V) \geq 2$

Domanda 4

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the quadratic form defined by

$$q(x, y) = (x, y)A\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

(a) The quadratic form $q(x, y)$ is negative definite.

HELP CENTER

liano (it)

Risposta corretta.

La risposta corretta è: $\mathcal{C}$ is a parabola.

Domanda 9

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$.

Which one of the following statements is true?

- (a) f has two distinct eigenvalues.
- (b) $(2, 0)$ is an eigenvector. ✓
- (c) $(0, 1)$ is an eigenvector.
- (d) f is simple.

Risposta corretta.

La risposta corretta è: $(2, 0)$ is an eigenvector.

Domanda 10

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Given the straight lines

$$r : \begin{cases} 2x - 5y = 7 \\ 2z - 3y = 5 \end{cases} \quad , \quad \text{and} \quad s : \begin{cases} x - y - z = 1 \\ x - 4y + z = 6, \end{cases}$$

find the right statement.

- (a) The lines are skew. ✗
- (b) The lines intersect at exactly one point.
- (c) None of the other statements is correct.
- (d) The lines are parallel and distinct.

Risposta errata.

La risposta corretta è: None of the other statements is correct.

HELP CENTER

liano (it)

La risposta corretta è: $\det(A \cdot B) = 5$.

Domanda 7
Risposta corretta
Punteggio ottenuto 100,00 su 100,00
Contrassegna domanda

If the linear system $AX = B$ with 4 equations and 5 variables has ∞^m solutions with $m \geq 1$, then necessarily

(a) $\mathrm{rank}(A|B) = 4$.
(b) $\mathrm{rank}(A) = 4$.
(c) $\mathrm{rank}(A) \geq 4$.
(d) $\mathrm{rank}(A) \leq 4$. ✓

Risposta corretta.
La risposta corretta è: $\mathrm{rank}(A) \leq 4$.

Domanda 8
Risposta corretta
Punteggio ottenuto 100,00 su 100,00
Contrassegna domanda

In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic $\mathcal{C}$ having equation
$x^2 - 6xy + 9y^2 + 2x - 4y = 0$.
Which of the following statements is true?

(a) $\mathcal{C}$ is a hyperbola.
(b) $\mathcal{C}$ is an ellipse.
(c) $\mathcal{C}$ is a parabola. ✓
(d) $\mathcal{C}$ is degenerate.

Risposta corretta.
La risposta corretta è: $\mathcal{C}$ is a parabola.

Domanda 9

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$

HELP CENTER

Italiano (it)

La risposta corretta è: $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

Domanda 5
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

Let $f : \mathbb{R}^4 \to \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - 3)$.

Which one of the following statements is true?

- (a) $dim(\text{Im}(f)) = 4$.
- (b) $dim(\text{Im}(f)) = 3$.
- (c) $dim(\text{Ker}(f)) = 2$. ✗
- (d) $dim(\text{Im}(f)) = 2$.

Risposta errata.

La risposta corretta è: $dim(\text{Im}(f)) = 4$.

Domanda 6
Risposta corretta
Punteggio ottenuto 100,00 su 100,00
Contrassegna domanda

Let $\cdot$ denote the usual matrix multiplication. If

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix},$$

then

- (a) $\det(\mathbf{A} \cdot \mathbf{B}) = 0$.
- (b) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$.
- (c) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$.
- (d) $\det(\mathbf{A} \cdot \mathbf{B}) = 5$. ✓

Risposta corretta

HELP CENTER

ano (it)

Domanda 4

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the quadratic form defined by

$$q(x,y) = (x,y)A\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

- (a) The quadratic form $q(x,y)$ is negative definite.
- (b) $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$. ✓
- (c) The matrix A has two eigenvalues of different sign.
- (d) $q(x,y) = 0$ is the equation of two lines passing through the origin.

Risposta corretta.

La risposta corretta è: $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

Domanda 5

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Let $f : \mathbb{R}^4 \to \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2-2)(t^2-3)$.

Which one of the following statements is true?

- (a) $dim(\text{Im}(f)) = 4$.
- (b) $dim(\text{Im}(f)) = 3$.
- (c) $dim(\text{Ker}(f)) = 2$. ✗
- (d) $dim(\text{Im}(f)) = 2$.

Risposta errata.

La risposta corretta è: $dim(\text{Im}(f)) = 4$.

Politecnico di Torino

Linear a

and geometry 1 - TTPU

7 8 9 10

Iniziato	venerdì, 4 febbraio 2022, 08:04
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Valutazione	**21,62** su un massimo di 33,00 (**66**%)

Domanda 1

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

In the euclidean 3-dimensional space with a fixed cartesian system, let us

$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = x + y = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2$

having radius respectively R_1 and R_2.

Which of the following statements is true?

- (a) $R_1 > R_2$.
- (b) $\mathcal{C}_1$ consists of only one point.
- (c) $R_1 < R_2$. ✗
- (d) $R_1 = R_2$.

Risposta errata.

La risposta corretta è: $R_1 = R_2$.

Domanda 2

In the euclidean space let us consider the points

HELP CENTER

Question 1

Not answered

Marked out of 100.00

Flag question

Let P, L, and U be the factors of the $PA = LU$ factorization of a given matrix A, and let $s = 2$ be the total number of performed row swaps. If

$$U = \begin{pmatrix} 2 & 5 & 7 \\ 0 & 1 & 9 \\ 0 & 0 & 3 \end{pmatrix}$$

then the determinant of A is equal to:

- 2
- 6
- -6
- -2

The correct answer is: 6

Question 2

Correct

Mark 100.00 out of 100.00

Flag question

Consider a floating point arithmetic with $N = 10$, $t = 3$ digit reserved for the mantissa and rounding technique "rounding to even". Let p be the mantissa and $\bar{p}$ the corresponding machine mantissa. If $p = 0.13819$, then:

(a) $\bar{p} = 0.1382$

(b) $\bar{p} = 0.138$ ✓

(c) $\bar{p} = 0.139$

HELP CENTER

Question 2

Correct

Mark 100.00 out of 100.00

Flag question

Consider a floating point arithmetic with $N = 10, t = 3$ digit reserved for the mantissa and rounding technique "rounding to even". Let p be the mantissa and $\bar{p}$ the corresponding machine mantissa. If $p = 0.13819$, then:

- (a) $\bar{p} = 0.1382$
- (b) $\bar{p} = 0.138$ ✓
- (c) $\bar{p} = 0.139$
- (d) $\bar{p} = 0.1381$

The correct answer is: $\bar{p} = 0.138$

Question 3

Not answered

Marked out of 100.00

Flag question

Given a matrix A of dimension $n \times n$, assume one applies to A the inverse power method, with the aim of finding the eigenvalue λ closest to $p = 2$ and the corresponding eigenvector x. Let wk be a MATLAB variable containing the approximation of x at the k-th iteration of the method.

Which one of the following MATLAB instructions creates a variable lambdak containing the approximation of λ at step k?

- zk = (A - 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);
- zk = (A + 2*eye(n))\wk; lambdak = 2 - 1/(wk'*zk);
- zk = (A + 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);
- zk = (A - 2*eye(n))\wk; lambdak = 2 - 1/(wk'*zk);

The correct answer is: zk = (A - 2*eye(n))\wk; lambdak = 2 + 1/(wk'*zk);

HELP CENTER

Mark -15.00 out of 100.00

Flag question

- A(:,end)
- col(10,A) ✗
- col(A,10)
- A('all',10)

The correct answer is: A(:,end)

Question 5

Not answered

Marked out of 100.00

Flag question

Let the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 4 & 5 \end{pmatrix}$$

be given, and assume to apply the power method to A. Let $w^{(m)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ be the approximation of the eigenvector corresponding to the eigenvalue of maximum modulus λ_1 at step m.

What is the value of $\lambda_1^{(m)}$, that is the approximation of λ_1 at step m?

- 15
- 9
- 3
- 8

HELP CENTER

The correct answer is: 3

Question 6
Not answered
Marked out of 100.00
Flag question

We are given a matrix $A \in R^{n,n}$, symmetric and positive definite, and its Choleski factorization $A = R^T R$. Which one of the following statements is surely correct?

- the factorization always exists but it is not unique
- the uniqueness of the factorization depends od the rank of matrix A
- the factorization does not always exist, but when it exists it is unique
- the factorization always exists and it is unique

The correct answer is: the factorization always exists and it is unique

Question 7
Incorrect
Mark -15.00 out of 100.00
Flag question

Consider the following set of Matlab commands:

```
x=linspace(-1,1,10);
xx=linspace(-1,1);
y=sin(pi*x);
z=spline(x,y,xx);
```

What is their result?

- They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$ ✕
- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$

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The correct answer is: the factorization always exists and it is unique

Question 7

Incorrect

Mark -15.00 out of 100.00

Flag question

Consider the following set of Matlab commands:

```
x=linspace(-1,1,10);
xx=linspace(-1,1);
y=sin(pi*x);
z=spline(x,y,xx);
```

What is their result?

- They compute the coefficients of the not-a-knot cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$. ✗
- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

The correct answer is: They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$

Question 8

Correct

Mark 100.00 out of 100.00

In exact arithmetic (base 10), the result of the multiplication

$$513.21 * 20.135$$

represented in the normalized floating point notation is:

HELP CENTER

spaced points in $[-1, 1]$.

- They compute the coefficients of the natural cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the constrained cubic spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.
- They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

The correct answer is: They compute and evaluate, at 100 equally spaced points in $[-1, 1]$, the not-a-knot spline interpolating the function $f(x) = \sin(\pi x)$ in 10 equally spaced points in $[-1, 1]$.

Question **8**

Correct

Mark 100.00 out of 100.00

Flag question

In exact arithmetic (base 10), the result of the multiplication

$513.21 * 20.135$

represented in the normalized floating point notation is:

- $10.33348337 * 10^3$
- $0.10333 * 10^6$
- $0.1033348335 * 10^5$ ✓
- $1.0333 * 10^4$

The correct answer is: $0.1033348335 * 10^5$

Finish review

HELP CENTER

Domanda 9

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

In the Euclidean plane with fixed cartesian system Oxy, let $\mathcal{C}$ be the conic defined by the equation $5x^2 + y^2 - 10y + 25 = 0$.

Which of the following statements is correct?

- (a) $\mathcal{C}$ is a parabola.
- (b) $\mathcal{C}$ is a hyperbola.
- (c) None of the other statements is correct.
- (d) $\mathcal{C}$ is a point. ✓

Risposta corretta.

La risposta corretta è: $\mathcal{C}$ is a point.

Domanda 10

Risposta non data

Punteggio max.: 100,00

Contrassegna domanda

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $\mathbb{R}_3[x]$ is a vector subspace?

- (a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.
- (b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.
- (c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$.
- (d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.

Risposta errata.

La risposta corretta è: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$

Fine revisione

HELP CENTER

La risposta corretta è: $\dim(\mathrm{Ker}\, f) = 1$.

Domanda 7
Risposta non data
Punteggio max.: 100,00
Contrassegna domanda

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1 : x + y + z = 0$ $\pi_2 : x + y + z = 1$ let us denote by R_i the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_2 = R_1$.
(b) C_1 and C_2 coincide.
(c) $R_2 < R_1$.
(d) $R_1 < R_2$.

Risposta errata.
La risposta corretta è: $R_2 < R_1$.

Domanda 8
Risposta corretta
Punteggio ottenuto 100,00 su 100,00
Contrassegna domanda

Let A be a real $n \times n$ matrix and let $A \cdot X = 0$ be a homogeneous linear system with n variables. Assume that the system has a non-zero solution and find the correct statement.

(a) $\det(A) = 0$. ✓
(b) $\mathrm{rank}(A) = n$.
(c) A is an invertible matrix.
(d) None of the previous statements is correct.

Risposta corretta.
La risposta corretta è: $\det(A) = 0$.

Domanda 9
Risposta corretta

In the Euclidean plane with fixed cartesian system Oxy, let C be the conic defined by the equation
$5x^2 + y^2 - 10y + 25 = 0$.

Italiano (it)

HELP CENTER

Risposta corretta.

La risposta corretta è: There exist infinitely many planes containing r and the point $(2, 2, 0)$

Domanda 6

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f : \mathbb{R}^3 \to \mathbb{R}_2[x]$ be a linear map such that

$$f(1,0,0) = x^2 + x, \quad f(0,1,0) = x + 2, \quad f(0,0,1) = 2x^2 + 4x + 4.$$

Which one of the following statements is true?

(a) $f(1, 0, -1) = x^2 - 4$.

(b) f is surjective.

(c) $\dim(\text{Ker } f) = 2$.

(d) $\dim(\text{Ker } f) = 1$. ✓

Risposta corretta.

La risposta corretta è: $\dim(\text{Ker } f) = 1$.

Domanda 7

Risposta non data

Punteggio max: 100,00

Contrassegna domanda

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the planes $\pi_1 : x + y + z = 0$ $\pi_2 : x + y + z = 1$ let us denote by R_i the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

(a) $R_2 = R_1$.

(b) C_1 and C_2 coincide.

(c) $R_2 < R_1$.

(d) $R_1 < R_2$.

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Risposta errata.

La risposta corretta è: The edges $\overline{P_1P_2}$ and $\overline{P_1P_3}$ are orthogonal for at least one value of t.

Domanda 5

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the line

$$r: \begin{cases} x = 2+t \\ y = 2-t \\ z = t \end{cases}$$

Which of the following statements is true?

(a) There exist infinitely many planes containing r and the point $(2, 2, 2)$.

(b) There exist infinitely many planes containing the point $(2, 2, 2)$ and orthogonal to r.

(c) There exist infinitely many planes containing r and the points $(2, 2, 2)$ and $(2, 2, 0)$.

(d) There exist infinitely many planes containing r and the point $(2, 2, 0)$. ✓

Risposta corretta.

La risposta corretta è: There exist infinitely many planes containing r and the point $(2, 2, 0)$.

Domanda 6

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f: \mathbb{R}^3 \to \mathbb{R}_2[x]$ be a linear map such that

$$f(1,0,0) = x^2 + x, \quad f(0,1,0) = x+2, \quad f(0,0,1) = 2x^2 + 4x + 4.$$

Which one of the following statements is true?

La risposta corretta è: None of the other statements is correct.

Domanda 4

Risposta non data

Punteggio max.: 100,00

Contrassegna domanda

For all $t \in \mathbb{R}$, consider the triangle having vertices in

$$P_1 = (3, 2, 1), \quad P_2 = (t, 1, 1), \quad P_3 = (1, 2, t).$$

Which of the following statements is true?

- (a) The edges $\overline{P_1P_2}$ and $\overline{P_2P_3}$ are orthogonal for $t = 3$.
- (b) There is no value of t such that the edges $\overline{P_1P_2}$ and $\overline{P_1P_3}$ are orthogonal.
- (c) The edges $\overline{P_1P_2}$ and $\overline{P_1P_3}$ are orthogonal for at least one value of t.
- (d) The edges $\overline{P_1P_2}$ and $\overline{P_2P_3}$ are orthogonal for two different values of t.

Risposta errata.

La risposta corretta è: The edges $\overline{P_1P_2}$ and $\overline{P_1P_3}$ are orthogonal for at least one value of t.

Domanda 5

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Let us consider the line

$$r : \begin{cases} x = 2 + t \\ y = 2 - t \\ z = t \end{cases}$$

Which of the following statements is true?

- (a) There exist infinitely many planes containing r and the point $(2, 2, 2)$.

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(d) $B \cdot A$ is skew-symmetric if and only if $b_1 = b_2 = b_3 = 0$. ✓

Risposta corretta.

La risposta corretta è: $B \cdot A$ is skew-symmetric if and only if $b_1 = b_2 = b_3 = 0$.

Domanda 3

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Consider the matrix $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 10 \end{pmatrix} \in \mathbb{R}^{3,3}$.

Find the right statement.

(a) A is positive definite. ✗

(b) A has negative eigenvalues.

(c) None of the other statements is correct.

(d) A does not have 0 as an eigenvalue.

Risposta errata.

La risposta corretta è: None of the other statements is correct.

Domanda 4

Risposta non data

Punteggio max.: 100,00

Contrassegna domanda

For all $t \in \mathbb{R}$, consider the triangle having vertices in

$$P_1 = (3, 2, 1), \quad P_2 = (t, 1, 1), \quad P_3 = (1, 2, t).$$

Which of the following statements is true?

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Punteggio 470,00/1000,00

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Domanda 1

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Let A a square matrix $n \times n$ with real coefficients such that $\det(A) \neq 0$ and $\det(A^2 + 2A) = 0$. Which of the following statements is correct?

(a) A has no real eigenvalue. ✗

(b) -2 is an eigenvalue for A.

(c) 0 is an eigenvalue for A.

(d) None of the other statements is correct.

Risposta errata.

La risposta corretta è: -2 is an eigenvalue for A

Domanda 2

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Consider the matrices

$$A = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

Find the correct statement.

(a) $B \cdot A$ is skew-symmetric for any $(b_1, b_2, b_3) \in \mathbb{R}^3$.

(b) There does not exist $(b_1, b_2, b_3) \in \mathbb{R}^3$ such that $B \cdot A$ is skew-symmetric.

(c) $B \cdot A$ is skew-symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is not orthogonal to $(1, 2, 3) \in \mathbb{R}^3$.

(d) $B \cdot A$ is skew-symmetric if and only if $b_1 = b_2 = b_3 = 0$. ✓

Risposta corretta.

La risposta corretta è: $B \cdot A$ is skew-symmetric if and only if $b_1 = b_2 = b_3 = 0$.

Didattica Italiano (it)

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

Find the right statement.

(a) 0 is an eigenvalue of A. ✗

(b) None of the other statements is correct.

(c) If $p_A(t)$ is the characteristic polynomial of A, then $p_A(t) > 0$ for infinite values of $t \in \mathbb{R}$.

(d) A is positive or negative definite.

Risposta errata.

La risposta corretta è: If $p_A(t)$ is the characteristic polynomial of A, then $p_A(t) > 0$ for infinite values of $t \in \mathbb{R}$.

Domanda 10

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Consider the following linear system with variables x_1, x_2, x_3, x_4:

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ x_2 + x_3 + x_4 = 0 \\ 2x_1 + x_2 + 3x_3 = 0 \\ x_1 - x_2 + x_3 - x_4 = 0 \end{cases}$$

Find the correct statement.

(a) The system has ∞^2 solutions.

(b) The system has exactly one solution.

(c) The system does not have solutions.

(d) The system has ∞^1 solutions. ✓

Risposta corretta.

La risposta corretta è: The system has ∞^1 solutions.

Fine revisione

Risposta errata.

La risposta corretta è: $A = \begin{pmatrix} 0 & 0 & 2 \\ -1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

Domanda 8
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

Consider the lines

$$r : \begin{cases} x = 1 \\ y = 3 \end{cases} \qquad s : \begin{cases} x = -2 \\ z = 3 \end{cases}$$

Which of the following statements is true?

- (a) There exists a plane containing r which is orthogonal to s.
- (b) There is no plane containing s which is orthogonal to r. ✗
- (c) There is no plane containing r which is parallel to s.
- (d) There exists a plane containing r and s.

Risposta errata.

La risposta corretta è: There exists a plane containing r which is orthogonal to s.

Domanda 9
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

Let A be a real 6×6 symmetrix matrix with zero trace and positive determinant.

Find the right statement.

- (a) 0 is an eigenvalue of A. ✗
- (b) None of the other statements is correct.

Didattica Italiano (it)

Domanda 7

Risposta non data

Punteggio max.: 100,00

Contrassegna domanda

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f : \mathbb{R}_2[x] \to \mathbb{R}_2[x]$ be the endomorphism defined as:

$$f(ax^2 + bx + c) = 2cx^2 - ax + 3b$$

and let A be the matrix associated to f with respect to the basis $\mathcal{B} = (x^2, x, 1)$ (for the domain and codomain).

Which one of the following statements is true?

(a) $A = \begin{pmatrix} 0 & 0 & -1 \\ 3 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$

(b) $A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 1 & 3 & 0 \end{pmatrix}$

(c) $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

(d) $A = \begin{pmatrix} 0 & 0 & 2 \\ -1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

Risposta errata.

La risposta corretta è: $A = \begin{pmatrix} 0 & 0 & 2 \\ -1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

Domanda 8

Risposta errata

Consider the lines

Didattica Italiano (it)

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

denote by R_i the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

- (a) $R_2 = R_1$.
- (b) C_1 and C_2 coincide.
- (c) $R_1 < R_2$.
- (d) $R_2 < R_1$. ✓

Risposta corretta.

La risposta corretta è: $R_2 < R_1$

Domanda 6

Risposta corretta

Punteggio ottenuto 100,00 su 100,00

Contrassegna domanda

Consider the matrices

$$A = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 & b_2 & b_3 \end{pmatrix}.$$

Find the correct statement.

- (a) $A \cdot B$ is symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is not parallel to $(1, 0, 3) \in \mathbb{R}^3$.
- (b) $A \cdot B$ is symmetric for any $(b_1, b_2, b_3) \in \mathbb{R}^3$.
- (c) There does not exist $(b_1, b_2, b_3) \in \mathbb{R}^3$ such that $A \cdot B$ is symmetric.
- (d) $A \cdot B$ is symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is parallel to $(1, 0, 3) \in \mathbb{R}^3$ ✓

Risposta corretta.

La risposta corretta è: $A \cdot B$ is symmetric if and only if $(b_1, b_2, b_3) \in \mathbb{R}^3$ is parallel to $(1, 0, 3) \in \mathbb{R}^3$.

Domanda 7

Risposta non data

Let $\mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2. Let $f : \mathbb{R}_2[x] \to \mathbb{R}_2[x]$ be the endomorphism defined as:

Didattica Italiano (it)

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

- (a) None of the other statements is correct.
- (b) A has no real eigenvalue. ✗
- (c) 0 is an eigenvalue for A.
- (d) -1 is an eigenvalue for A.

Risposta errata.

La risposta corretta è: -1 is an eigenvalue for A

Domanda 4

Risposta errata

Punteggio ottenuto -15,00 su 100,00

Contrassegna domanda

For all $t \in \mathbb{R}$, let us consider the points

$$P = (2,2,2), \quad Q = (0,0,t), \quad R = (1,-1,t), \quad S = (-3,2,1+t).$$

Which of the following statements is true?

- (a) The points P, Q, R, S are vertices of a trapezoid for all values of t. ✗
- (b) The points P, Q, R, S are coplanar when $t = 7$.
- (c) The points P, Q, R, S are vertices of a rectangle for all values of t.
- (d) The points P, Q, R, S are coplanar for at least one value of $t \in \mathbb{R}$.

Risposta errata.

La risposta corretta è: The points P, Q, R, S are coplanar for at least one value of $t \in \mathbb{R}$.

Domanda 5

Risposta corretta

Punteggio ottenuto

Given the sphere $S : x^2 + y^2 + z^2 - 2x + 2z = 14$, and the planes $\pi_1 : x + y + z = 0$ $\pi_2 : x + y + z = 3$ let us denote by R_i the radius of the circle $S \cap \pi_i$ and by C_i its center, then:

Didattica   Italiano (it)

Help Center

Domanda 1
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

Let $h \in \mathbb{R}$ and let U be the vector subspace of $\mathbb{R}^3$ generated by the vector $(0, 0, h)$.
Let $V = \{(x, y, z) \in \mathbb{R}^3 : x + 3y + z = 0\}$.
Which one of the following statements is true?

- (a) For $h = 0$ it holds that $U \cap V = \emptyset$.
- (b) $\dim(U \cap V) = 1$ for any h.
- (c) $\dim(U) = 1$ for any h. ✗
- (d) $\dim(U \cap V) = 0$ for any h.

Risposta errata.
La risposta corretta è: $\dim(U \cap V) = 0$ for any h

Domanda 2
Risposta errata
Punteggio ottenuto -15,00 su 100,00
Contrassegna domanda

In the Euclidean plane with fixed cartesian system Oxy, let C be the conic defined by the equation
$-x^2 + 9y^2 + x + 3y = 0$.
Which of the following statements is correct?

- (a) C is the union of two non parallel lines.
- (b) None of the other statements is correct. ✗
- (c) C is a parabola.
- (d) C is an ellipse.

Risposta errata.
La risposta corretta è: C is the union of two non parallel lines.

Domanda 3
Risposta errata

Let A a square matrix $n \times n$ with real coefficients such that $\det(A) \neq 0$ and $\det(A^2 + A) = 0$. Which of the following statements is correct?

... 1 - TTPU

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Domanda 2

Risposta non ancora data

Punteggio max.: 100,00

Contrassegna domanda

Consider the sphere $S : x^2 + y^2 + z^2 - 2x + 2y = 7$, and the plane $\pi : x + y + z = 0$.
The circle $\pi \cap S$ has radius R and center C such that

- (a) $C = (1, -1, 0)$, $R = 3$
- (b) $C = (0, 0, 0)$, $R = 3$
- (c) $C = (1, -2, 1)$, $R = 7$
- (d) $C = (-1, 1, 0)$, $R = 3$

Precedente

Successivo

HELP CENTER

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the followin subsets of $\mathbb{R}_3[x]$ is a vector subspace?

(a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

(b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗

(c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.

(d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let $\mathcal{C}$ be the conic defined by the equation

$$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0.$$

Which of the following statements is correct?

(a) $\mathcal{C}$ is a hyperbola.

(b) $\mathcal{C}$ is degenerate. ✗

(c) $\mathcal{C}$ is a parabola.

(d) $\mathcal{C}$ is an ellipse.

Risposta errata.

The correct answer is: $\mathcal{C}$ is a parabola.

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(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$

Risposta corretta.

The correct answer is: $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$

Question 9

Incorrect

Mark -15.00 out of 100.00

Flag question

Let $\mathbb{R}_3[x]$ be the vector space of polynomials in the variable x, of degree less or equal than 3 and with coefficients in $\mathbb{R}$. Which one of the following subsets of $\mathbb{R}_3[x]$ is a vector subspace?

(a) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

(b) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ad - bc = 0\}$. ✗

(c) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b + c + d = 1\}$.

(d) The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : ac = 0\}$.

Risposta errata.

The correct answer is: The set $\{p(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}_3[x] : a + b = 0\}$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Flag question

In the Euclidean plane with fixed cartesian system Oxy, let C be the conic defined by the equation

$x^2 - 2xy + y^2 + 3x + 3y + 1 = 0$.

Which of the following statements is correct?

(a) C is a hyperbola.

Risposta errata.

The correct answer is: f is simple.

Question 8

Correct

Mark 100.00 out of 100.00

Flag question

Consider the matrix

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & -1 & 2 \end{pmatrix}$$

Find the correct statement.

(a) $A^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & -1 & -3 \\ 1 & -1 & -1 \end{pmatrix}$

(b) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(c) $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 2 & 2 \\ -1 & 1 & 1 \end{pmatrix}$

(d) $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 2 & -3 \\ 1 & 1 & -1 \end{pmatrix}$ ✓